

SECTOR ROTATION REPORT

Macro-Regime Equity Rotation

This report isolates the sector rotation layer from the broader macro atlas. It focuses on which equity types have historically held up best, made higher highs most often, and offered the strongest 1-year to 3-year forward return profile inside similar macro regimes.

Generated 2026-05-09 14:13 UTC

Current regime: Fragile Late-Cycle Watch

Quadrant: Growth Down / Inflation Down

Cash rule: 40.0%

Top entry: Information Technology • 1Y +16.7% • confidence High

METHOD

The report reuses the same monthly macro regime engine as the macro atlas, then scores sector ETF proxies by 1-year and 3-year forward returns, future drawdown, and higher-high frequency inside matching regimes. No point-in-time analyst EPS feed exists in this repo. The sector entry calls below therefore use earnings-sensitivity proxies plus realized 1-year and 3-year forward returns from historically similar macro regimes.

CURRENT OUTLOOK

What The Rotation Layer Is Saying Now

CURRENT MACRO REGIME

Fragile Late-Cycle Watch

Growth is soft but not broken.
Inflation is near balance. Valuation is rich. Credit is mixed rather than broken.

GROWTH / INFLATION QUADRANT

Growth Down / Inflation Down

A slowdown without inflation pressure can become a recovery setup if stress variables ease, but credit and volatility have to confirm it.

PORTFOLIO STRUCTURE

Cash 40.0%

The cash sleeve is fixed so the model only rotates the 60% equity bucket according to the active regime.

MOST PROBABLE ENTRY

Information Technology

Expected 1Y +16.7%, expected 3Y +15.1%, higher-high hit rate 90%, confidence 93.

DEFENSIVE SLEEVE

Consumer Staples

Defensive sleeve candidate with portfolio weight 13.5% and average future drawdown -2.3%.

ML OVERLAY

Machine Learning Ensemble Audit

Features are lagged by one bar and each validation window uses a five-bar purge plus a five-bar embargo. The sector ML study now trains from 2000 onward so the walk-forward history can include 2008, 2020, and 2022 while preserving a separate untouched 2025+ holdout. Macro regime attribution comes from the local macro store and is useful for regime slicing, but it is not a point-in-time macro vintage database.

VALIDATION DESIGN

**2000-2024 walk-forward,
2025+ holdout**

Five-year expanding train windows, one-year validation windows, feature lag 1, purge 5 bars, embargo 5 bars. The broader benchmark section starts in 2006 so crisis eras are visible.

MODEL DESIGN

Probability averaging only

Elastic Net, ExtraTrees, and LGBMClassifier are averaged directly. The stacked gate is disabled in this study to reduce meta-model overfit risk.

MOST STABLE MODEL

Average Ensemble

Won 3 sectors on the stability score that penalizes validation-to-holdout degradation.

HOLDOUT LEADER

Utilities

Best ensemble holdout by Sharpe. CAGR +16.7%, Sharpe 2.87, max drawdown -3.2%.

COST ROBUSTNESS

6 / 8 sectors

Still positive at the worst slippage stress scenario inside the untouched holdout.

LIVE ROTATION

Current ML-Adjusted Rotation

This allocation blends the current regime prior with the latest complete ML holdout signal and validation-derived sector quality.

TIMING

TOP COMBINED PICK

Latest complete ML signal:
2026-03-03

The ML leg uses the latest fully labeled holdout signal. The historical benchmark below uses only these untouched holdout dates.

Utilities

Combined score 0.575, live probability 60%, portfolio weight 60.0%.

SECTOR	REGIME SCORE	ML PROBABILITY	VALIDATION QUALITY	STABILITY	COMBINED SCORE
Utilities (XLU)	0.481	60%	0.125	98.8	0.575

HOLDOUT BENCHMARK

Strict Holdout: ML Rotation, Reserve Cash Rule, And SPY

This benchmark uses only the final untouched holdout dates after the validation years. Signals are generated from walk-forward out-of-sample sector predictions, executed one bar after the signal date, and held for 5 bars before the next rebalance. The quality-weighted variant mixes live ensemble probability with a sector quality prior estimated only from the pre-holdout validation window. The reserve-cash rule keeps 40% of the portfolio in cash earning 5% annually, deploys 10% of that reserve after a 5% SPY drawdown, another 10% after a 10% drawdown, and the rest after a 20% drawdown, then rebuilds the reserve only after SPY gets back to its prior high. The leveraged scenario applies 3x gross exposure and charges 6% annual interest on the borrowed 2x capital over each realized holding period. Reserve Cash Rule x3 Drawdown Sleeve @ 6% only levers the deployed reserve sleeve during drawdowns and rotates that sleeve back to cash once SPY returns to a fresh high.

BENCHMARK SCOPE

Untouched 2025+ holdout

STRATEGY

ML Quality-Weighted Rotation

This slice uses only the final untouched holdout dates after the validation years. Window range 2025-01-02 to 2026-03-11 across 30 realized five-bar holding windows.

CAGR +11.0%, Sharpe 1.14, max drawdown -3.0%, turnover 20.7/yr.

BENCHMARK

SPY Buy And Hold

CAGR +11.1%, Sharpe 1.30, max drawdown -2.8% over the same benchmark windows.

RESERVE RULE

Sector Reserve Cash Rule

CAGR +8.3%, Sharpe 1.39, max drawdown -1.6%, turnover 0.1/yr.

LEVERAGED RESERVE

Reserve Cash Rule x3 Drawdown Sleeve @ 6%

CAGR +8.6%, Sharpe 1.40, max drawdown -1.6%, turnover 0.4/yr. Only the deployed reserve sleeve is levered.

TRIGGER DEPTH

Worst SPY signal drawdown -8.2%

Peak reserve deployment reached 10.0% of the reserve sleeve. Only the first 5% drawdown tier triggered in this slice.

LEVERAGED STRATEGY

ML Quality-Weighted Rotation x3 @ 6%

CAGR +25.1%, Sharpe 0.92, max drawdown -10.4%, turnover 20.7/yr.

LEVERAGED BENCHMARK

SPY Buy And Hold x3 @ 6%

CAGR +26.1%, Sharpe 1.05, max drawdown -10.2% under the same 6% financing assumption.

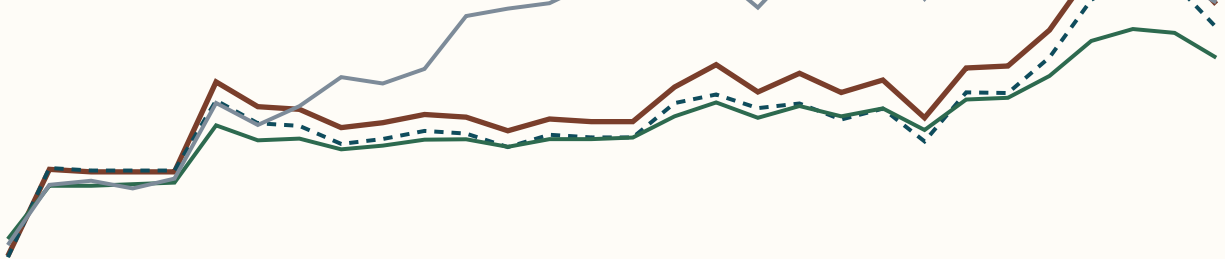
METRIC DEFINITION

Latest complete signal 2026-03-03

Windows count realized 5-bar evaluation periods. Trades count actual entries or rebalances. SPY buy-and-hold therefore shows 1 entry across 30 evaluation windows.

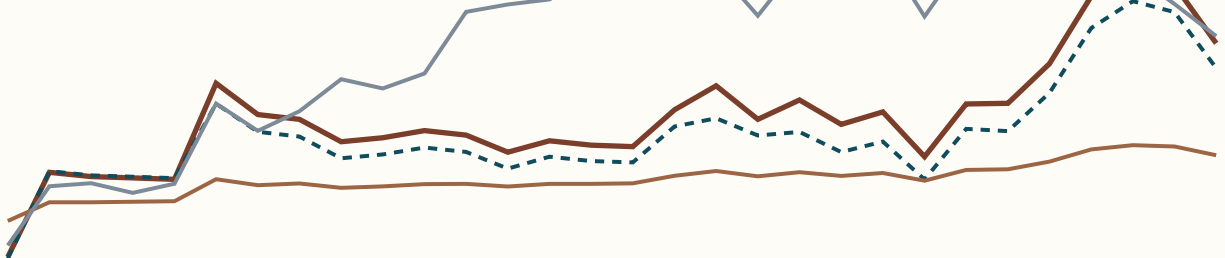
Holdout Equity Curves

ML quality-weighted: +12.9% | ML probability-only: +11.5% | Reserve cash rule: +9.6% | SPY: +13.0%



Holdout Equity Curves x3 @ 6% Financing

ML quality-weighted x3: +29.5% | ML probability-only x3: +25.2% | Reserve drawdown sleeve x3: +10.0% | SPY x3: +30.7%



STRATEGY	TOTAL RETURN	CAGR	SHARPE	SORTINO	MAX DD	CALMAR
ML Probability Rotation	+11.5%	+9.8%	1.09	2.39	-2.7%	3.64
ML Quality-Weighted Rotation	+12.9%	+11.0%	1.14	2.61	-3.0%	3.73
Sector Reserve	+9.6%	+8.3%	1.39	4.00	-1.6%	5.29

STRATEGY	TOTAL RETURN	CAGR	SHARPE	SORTINO	MAX DD	CALMAR
Cash Rule						
Reserve Cash Rule x3 Drawdown Sleeve @ 6%	+10.0%	+8.6%	1.40	4.19	-1.6%	5.50
SPY Buy And Hold	+13.0%	+11.1%	1.30	3.42	-2.8%	4.04
ML Probability Rotation x3 @ 6%	+25.2%	+21.4%	0.85	1.86	-11.1%	1.94
ML Quality- Weighted Rotation x3 @ 6%	+29.5%	+25.1%	0.92	2.03	-10.4%	2.41
SPY Buy And Hold x3 @ 6%	+30.7%	+26.1%	1.05	2.75	-10.2%	2.56

WALK-FORWARD HISTORY

Crisis-Inclusive History: ML Rotation, Reserve Cash Rule, And SPY

This benchmark uses every walk-forward out-of-sample window from 2006 onward, including the 2008 financial crisis, the 2020 shock, and the 2022 rate-reset drawdown.

Signals are generated from walk-forward out-of-sample sector predictions, executed one bar after the signal date, and held for 5 bars before the next rebalance. The quality-weighted variant mixes live ensemble probability with a sector quality prior estimated only from the pre-holdout validation window. The reserve-cash rule keeps 40% of the portfolio in cash earning 5% annually, deploys 10% of that reserve after a 5% SPY drawdown, another 10% after a 10% drawdown, and the rest after a 20% drawdown, then rebuilds the reserve only after SPY gets back to its prior high. The leveraged scenario applies 3x gross exposure and charges 6% annual interest on the borrowed 2x capital over each realized holding period. Reserve Cash Rule x3 Drawdown Sleeve @ 6% only levers the deployed reserve sleeve during drawdowns and rotates that sleeve back to cash once SPY returns to a fresh high.

BENCHMARK SCOPE

2006-2026 walk-forward history

This slice uses every walk-forward out-of-sample window from 2006 onward, so the major crisis regimes are visible. Window range 2006-01-03 to 2026-03-05 across 870 realized five-bar holding windows.

STRATEGY

ML Quality-Weighted Rotation

CAGR +11.2%, Sharpe 0.82, max drawdown -35.6%, turnover 31.3/yr.

BENCHMARK

SPY Buy And Hold

CAGR +8.1%, Sharpe 0.58, max drawdown -50.3% over the same benchmark windows.

RESERVE RULE

Sector Reserve Cash Rule

CAGR +10.0%, Sharpe 0.84, max drawdown -31.5%, turnover 5.6/yr.

LEVERAGED RESERVE

Reserve Cash Rule x3 Drawdown Sleeve @ 6%

CAGR +10.3%, Sharpe 0.58, max drawdown -49.9%, turnover 16.9/yr.

TRIGGER DEPTH

Worst SPY signal drawdown -53.9%

Peak reserve deployment reached 100.0% of the reserve sleeve.

Only the deployed reserve sleeve is levered.

Multiple reserve tiers triggered in this slice.

LEVERAGED STRATEGY

ML Quality-Weighted Rotation x3 @ 6%

CAGR +16.6%, Sharpe 0.58, max drawdown -83.6%, turnover 31.3/yr.

LEVERAGED BENCHMARK

SPY Buy And Hold x3 @ 6%

CAGR +5.4%, Sharpe 0.36, max drawdown -93.4% under the same 6% financing assumption.

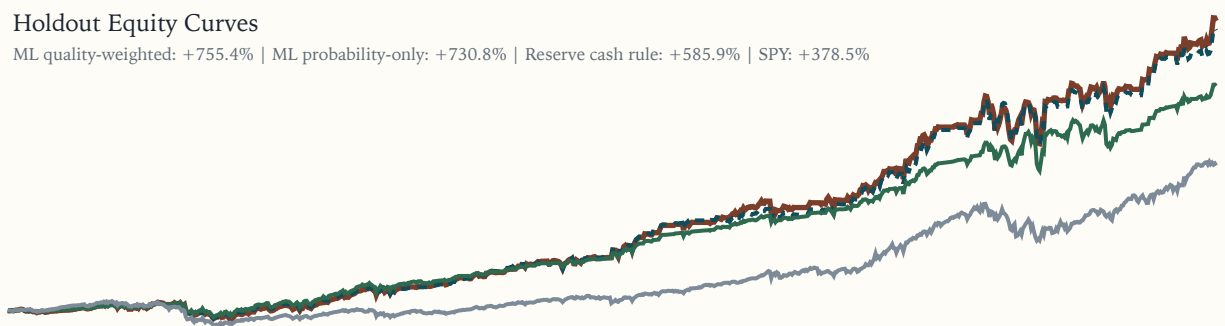
METRIC DEFINITION

Latest complete signal 2026-03-03

Windows count realized 5-bar evaluation periods. Trades count actual entries or rebalances. SPY buy-and-hold therefore shows 1 entry across 870 evaluation windows.

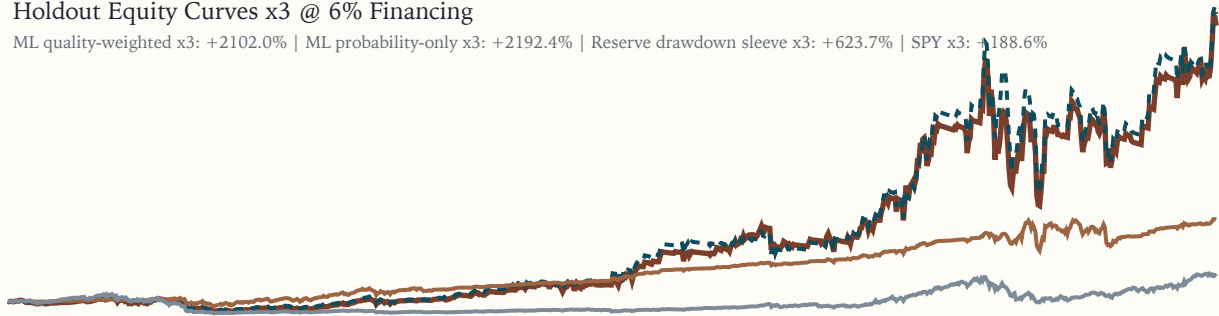
Holdout Equity Curves

ML quality-weighted: +755.4% | ML probability-only: +730.8% | Reserve cash rule: +585.9% | SPY: +378.5%



Holdout Equity Curves x3 @ 6% Financing

ML quality-weighted x3: +2102.0% | ML probability-only x3: +2192.4% | Reserve drawdown sleeve x3: +623.7% | SPY x3: +188.6%



STRATEGY	TOTAL RETURN	CAGR	SHARPE	SORTINO	MAX DD	CALMAR
ML Probability Rotation	+730.8%	+11.1%	0.85	1.02	-34.0%	0.33
ML Quality- Weighted Rotation	+755.4%	+11.2%	0.82	0.97	-35.6%	0.32
Sector Reserve Cash Rule	+585.9%	+10.0%	0.84	0.93	-31.5%	0.32
Reserve Cash Rule x3 Drawdown Sleeve @ 6%	+623.7%	+10.3%	0.58	0.61	-49.9%	0.21
SPY Buy And Hold	+378.5%	+8.1%	0.58	0.69	-50.3%	0.16

STRATEGY	TOTAL RETURN	CAGR	SHARPE	SORTINO	MAX DD	CALMAR
ML Probability Rotation x3 @ 6%	+2192.4%	+16.8%	0.59	0.77	-81.4%	0.21
ML Quality- Weighted Rotation x3 @ 6%	+2102.0%	+16.6%	0.58	0.75	-83.6%	0.20
SPY Buy And Hold x3 @ 6%	+188.6%	+5.4%	0.36	0.43	-93.4%	0.06

EXECUTION SENSITIVITY

How 5, 10, And 21-Bar Rebalancing Change The Result

This section uses the full 2006-2026 walk-forward history and compares the same sector signal under slower execution cadences. It answers whether lower turnover can preserve enough of the edge to matter in practice.

METHOD

Cadence overlay, not retraining

The cadence test reuses the same daily ML signal and only changes execution frequency to 5, 10, or 21 bars. This isolates turnover and holding-period effects without

QUALITY CADENCE

Quality best at 5 bars

Sharpe 0.82, CAGR +11.2%, max drawdown -35.6%, turnover 31.3/yr.

pretending the model was retrained for monthly horizons.

RESERVE CADENCE

Reserve rule best at 5 bars

Sharpe 0.84, CAGR +10.0%, max drawdown -31.5%, turnover 5.6/yr.

TURNOVER TRADEOFF

Lowest-turnover quality cadence: 21 bars

Turnover drops to 8.7/yr with CAGR +6.5% and max drawdown -37.4%.

BENCHMARK CADENCE

SPY strongest at 21 bars

CAGR +9.4%, Sharpe 0.75, max drawdown -48.6%.



CADENCE	STRATEGY	TOTAL RETURN	CAGR	SHARPE	MAX DD	TRADES
5-bar	ML Probability Rotation	+730.8%	+11.1%	0.85	-34.0%	683
5-bar	ML Probability	+2192.4%	+16.8%	0.59	-81.4%	683

CADENCE	STRATEGY	TOTAL RETURN	CAGR	SHARPE	MAX DD	TRADES
	Rotation x3 @ 6%					
5-bar	ML Quality- Weighted Rotation	+755.4%	+11.2%	0.82	-35.6%	683
5-bar	ML Quality- Weighted Rotation x3 @ 6%	+2102.0%	+16.6%	0.58	-83.6%	683
5-bar	Reserve Cash Rule x3 Drawdown Sleeve @ 6%	+623.7%	+10.3%	0.58	-49.9%	246
5-bar	SPY Buy And Hold	+378.5%	+8.1%	0.58	-50.3%	1
5-bar	SPY Buy And Hold x3 @ 6%	+188.6%	+5.4%	0.36	-93.4%	1
5-bar	Sector Reserve Cash Rule	+585.9%	+10.0%	0.84	-31.5%	246
10-bar	ML Probability Rotation	+643.4%	+10.5%	0.86	-36.8%	349

CADENCE	STRATEGY	TOTAL RETURN	CAGR	SHARPE	MAX DD	TRADES
10-bar	ML Probability Rotation x3 @ 6%	+1693.1%	+15.4%	0.59	-82.8%	349
10-bar	ML Quality- Weighted Rotation	+579.9%	+10.0%	0.77	-39.5%	349
10-bar	ML Quality- Weighted Rotation x3 @ 6%	+1034.7%	+12.8%	0.52	-87.8%	349
10-bar	Reserve Cash Rule x3 Drawdown Sleeve @ 6%	+238.9%	+6.3%	0.40	-58.8%	139
10-bar	SPY Buy And Hold	+380.3%	+8.1%	0.62	-51.8%	1
10-bar	SPY Buy And Hold x3 @ 6%	+198.1%	+5.6%	0.38	-94.5%	1
10-bar	Sector Reserve Cash Rule	+388.0%	+8.2%	0.72	-36.8%	139

CADENCE	STRATEGY	TOTAL RETURN	CAGR	SHARPE	MAX DD	TRADES
21-bar	ML Probability Rotation	+179.5%	+5.2%	0.48	-33.9%	168
21-bar	ML Probability Rotation x3 @ 6%	-3.2%	-0.2%	0.20	-81.8%	168
21-bar	ML Quality- Weighted Rotation	+254.8%	+6.5%	0.54	-37.4%	168
21-bar	ML Quality- Weighted Rotation x3 @ 6%	+79.2%	+2.9%	0.28	-85.0%	168
21-bar	Reserve Cash Rule x3 Drawdown Sleeve @ 6%	+224.4%	+6.0%	0.39	-50.3%	92
21-bar	SPY Buy And Hold	+508.9%	+9.4%	0.75	-48.6%	1
21-bar	SPY Buy And Hold x3 @ 6%	+756.4%	+11.2%	0.49	-92.7%	1

CADENCE	STRATEGY	TOTAL RETURN	CAGR	SHARPE	MAX DD	TRADES
21-bar	Sector Reserve Cash Rule	+268.7%	+6.7%	0.61	-31.6%	92

BENCHMARK SLICES

Holdout Performance By Year And Signal Regime

These slices keep only the strict 2025+ holdout windows and break them down by calendar year and signal regime.

STRATEGY	YEAR	TOTAL RETURN	CAGR	SHARPE	MAX DD	TRADES	W
ML Probability Rotation	2025	+6.8%	+7.2%	0.91	-2.7%	15	20
ML Probability Rotation x3 @ 6%	2025	+14.0%	+14.8%	0.69	-9.5%	15	20
ML Quality- Weighted Rotation	2025	+8.7%	+9.1%	1.04	-2.8%	15	20
ML Quality- Weighted Rotation x3 @ 6%	2025	+19.6%	+20.7%	0.84	-9.8%	15	20

STRATEGY	YEAR	TOTAL RETURN	CAGR	SHARPE	MAX DD	TRADES	W
Reserve Cash Rule x3 Drawdown Sleeve @ 6%	2025	+7.0%	+7.4%	1.31	-1.4%	3	20
SPY Buy And Hold	2025	+15.3%	+16.2%	2.01	-2.3%	15	20
SPY Buy And Hold x3 @ 6%	2025	+43.4%	+46.1%	1.78	-7.5%	15	20
Sector Reserve Cash Rule	2025	+6.7%	+7.0%	1.30	-1.4%	3	20
ML Probability Rotation	2026	+4.3%	n/a	n/a	-2.7%	8	10
ML Probability Rotation x3 @ 6%	2026	+9.7%	n/a	n/a	-8.5%	8	10
ML Quality- Weighted Rotation	2026	+3.9%	n/a	n/a	-2.7%	8	10
ML Quality- Weighted	2026	+8.2%	n/a	n/a	-8.5%	8	10

STRATEGY	YEAR	TOTAL RETURN	CAGR	SHARPE	MAX DD	TRADES	W
Rotation x3 @ 6%							
Reserve Cash Rule x3 Drawdown Sleeve @ 6%	2026	+2.8%	n/a	n/a	-1.6%	0	10
SPY Buy And Hold	2026	-2.0%	n/a	n/a	-2.8%	8	10
SPY Buy And Hold x3 @ 6%	2026	-8.8%	n/a	n/a	-10.2%	8	10
Sector Reserve Cash Rule	2026	+2.8%	n/a	n/a	-1.6%	0	10

STRATEGY	SIGNAL REGIME	TOTAL RETURN	CAGR	SHARPE	MAX DD	HIT RATE	T
ML Probability Rotation	Cash	-0.6%	n/a	n/a	-0.4%	0%	4
ML Probability Rotation x3 @ 6%	Cash	-3.4%	n/a	n/a	-2.7%	0%	4
ML Quality-	Cash	-0.6%	n/a	n/a	-0.4%	0%	4

STRATEGY	SIGNAL REGIME	TOTAL RETURN	CAGR	SHARPE	MAX DD	HIT RATE	T
Weighted Rotation							
ML Quality- Weighted Rotation x3 @ 6%	Cash	-3.4%	n/a	n/a	-2.7%	0%	4
Reserve Cash Rule x3 Drawdown Sleeve @ 6%	Cash	+0.6%	n/a	n/a	+0.0%	100%	0
SPY Buy And Hold	Cash	+6.7%	n/a	n/a	-0.5%	86%	4
SPY Buy And Hold x3 @ 6%	Cash	+19.0%	n/a	n/a	-1.6%	86%	4
Sector Reserve Cash Rule	Cash	+0.4%	n/a	n/a	+0.0%	100%	0
ML Probability Rotation	Fragile Late- Cycle Watch	+1.7%	n/a	n/a	-2.4%	50%	4
ML Probability	Fragile Late-	+3.0%	n/a	n/a	-8.2%	50%	4

STRATEGY	SIGNAL REGIME	TOTAL RETURN	CAGR	SHARPE	MAX DD	HIT RATE	T
Rotation x3 @ 6%	Cycle Watch						
ML Quality- Weighted Rotation	Fragile Late- Cycle Watch	+2.7%	n/a	n/a	-2.5%	50%	4
ML Quality- Weighted Rotation x3 @ 6%	Fragile Late- Cycle Watch	+5.9%	n/a	n/a	-8.5%	50%	4
Reserve Cash Rule x3 Drawdown Sleeve @ 6%	Fragile Late- Cycle Watch	+2.2%	n/a	n/a	-1.7%	50%	3
SPY Buy And Hold	Fragile Late- Cycle Watch	+5.8%	n/a	n/a	-1.2%	67%	4
SPY Buy And Hold x3 @ 6%	Fragile Late- Cycle Watch	+15.9%	n/a	n/a	-4.0%	67%	4
Sector Reserve Cash Rule	Fragile Late- Cycle Watch	+2.0%	n/a	n/a	-1.5%	50%	3

STRATEGY	SIGNAL REGIME	TOTAL RETURN	CAGR	SHARPE	MAX DD	HIT RATE	T
ML Probability Rotation	Liquidity Bubble Or Valuation Stretch	+10.3%	+8.8%	1.12	-2.7%	59%	1
ML Probability Rotation x3 @ 6%	Liquidity Bubble Or Valuation Stretch	+25.8%	+22.0%	0.96	-9.1%	59%	1
ML Quality-Weighted Rotation	Liquidity Bubble Or Valuation Stretch	+10.5%	+9.0%	1.13	-3.0%	65%	1
ML Quality-Weighted Rotation x3 @ 6%	Liquidity Bubble Or Valuation Stretch	+26.5%	+22.6%	0.97	-10.1%	65%	1
Reserve Cash Rule x3 Drawdown Sleeve @ 6%	Liquidity Bubble Or Valuation Stretch	+7.0%	+6.0%	1.25	-1.6%	65%	0
SPY Buy And Hold	Liquidity Bubble Or	+0.1%	+0.1%	0.05	-2.8%	47%	1

STRATEGY	SIGNAL REGIME	TOTAL RETURN	CAGR	SHARPE	MAX DD	HIT RATE	TRADES
Valuation Stretch							
SPY Buy And Hold x3 @ 6%	Liquidity Bubble Or Valuation Stretch	-5.2%	-4.5%	-0.14	-10.2%	47%	1
Sector Reserve Cash Rule	Liquidity Bubble Or Valuation Stretch	+7.0%	+6.0%	1.25	-1.6%	65%	0

HISTORY SLICES

2006-2026 Walk-Forward Performance By Year And Signal Regime

These slices use the full walk-forward out-of-sample history, so the year and regime tables include the major stress periods that are absent from the strict holdout alone.

STRATEGY	YEAR	TOTAL RETURN	CAGR	SHARPE	MAX DD	TRADES
ML Probability Rotation	2006	+5.5%	+5.7%	0.70	-8.0%	42

STRATEGY	YEAR	TOTAL RETURN	CAGR	SHARPE	MAX DD	TRADES
ML Probability Rotation x3 @ 6%	2006	+2.4%	+2.5%	0.22	-24.9%	42
ML Quality- Weighted Rotation	2006	+3.6%	+3.8%	0.46	-8.4%	42
ML Quality- Weighted Rotation x3 @ 6%	2006	-3.3%	-3.4%	0.01	-26.2%	42
Reserve Cash Rule x3 Drawdown Sleeve @ 6%	2006	+4.4%	+4.6%	0.66	-6.2%	21
SPY Buy And Hold	2006	+13.8%	+14.4%	1.54	-7.4%	42
SPY Buy And Hold x3 @ 6%	2006	+28.0%	+29.3%	1.09	-22.2%	42
Sector Reserve Cash Rule	2006	+4.4%	+4.5%	0.78	-5.2%	21

STRATEGY	YEAR	TOTAL RETURN	CAGR	SHARPE	MAX DD	TRADES
ML Probability Rotation	2007	+6.5%	+6.8%	0.55	-8.9%	44
ML Probability Rotation x3 @ 6%	2007	+2.7%	+2.8%	0.28	-30.5%	44
ML Quality- Weighted Rotation	2007	+7.9%	+8.2%	0.56	-12.3%	44
ML Quality- Weighted Rotation x3 @ 6%	2007	+3.9%	+4.0%	0.34	-37.9%	44
Reserve Cash Rule x3 Drawdown Sleeve @ 6%	2007	+7.8%	+8.1%	0.77	-7.2%	11
SPY Buy And Hold	2007	+2.2%	+2.3%	0.23	-11.8%	44
SPY Buy And Hold x3 @ 6%	2007	-10.1%	-10.4%	-0.02	-36.0%	44

STRATEGY	YEAR	TOTAL RETURN	CAGR	SHARPE	MAX DD	TRADES
Sector Reserve Cash Rule	2007	+7.2%	+7.4%	0.76	-7.0%	11
ML Probability Rotation	2008	-16.3%	-16.6%	-0.73	-26.3%	37
ML Probability Rotation x3 @ 6%	2008	-55.3%	-56.1%	-0.91	-66.8%	37
ML Quality- Weighted Rotation	2008	-15.1%	-15.4%	-0.63	-26.8%	37
ML Quality- Weighted Rotation x3 @ 6%	2008	-54.1%	-54.8%	-0.80	-67.6%	37
Reserve Cash Rule x3 Drawdown Sleeve @ 6%	2008	-18.4%	-18.8%	-0.42	-36.7%	21
SPY Buy And Hold	2008	-35.0%	-35.6%	-1.39	-42.3%	37

STRATEGY	YEAR	TOTAL RETURN	CAGR	SHARPE	MAX DD	TRADES
SPY Buy And Hold x3 @ 6%	2008	-82.2%	-82.8%	-1.52	-86.2%	37
Sector Reserve Cash Rule	2008	-10.4%	-10.6%	-0.45	-23.1%	21
ML Probability Rotation	2009	+17.8%	+18.1%	0.86	-20.0%	44
ML Probability Rotation x3 @ 6%	2009	+25.7%	+26.2%	0.69	-52.3%	44
ML Quality- Weighted Rotation	2009	+13.1%	+13.3%	0.63	-21.7%	44
ML Quality- Weighted Rotation x3 @ 6%	2009	+7.6%	+7.7%	0.48	-59.7%	44
Reserve Cash Rule x3 Drawdown Sleeve @ 6%	2009	+26.1%	+26.6%	0.74	-38.0%	23

STRATEGY	YEAR	TOTAL RETURN	CAGR	SHARPE	MAX DD	TRADES
SPY Buy And Hold	2009	+9.0%	+9.2%	0.49	-22.7%	44
SPY Buy And Hold x3 @ 6%	2009	-2.2%	-2.2%	0.33	-57.9%	44
Sector Reserve Cash Rule	2009	+17.2%	+17.5%	0.76	-21.9%	23
ML Probability Rotation	2010	+19.4%	+20.0%	1.13	-12.4%	49
ML Probability Rotation x3 @ 6%	2010	+38.0%	+39.3%	0.90	-36.8%	49
ML Quality- Weighted Rotation	2010	+21.0%	+21.7%	1.11	-12.4%	49
ML Quality- Weighted Rotation x3 @ 6%	2010	+40.7%	+42.1%	0.90	-36.8%	49
Reserve Cash Rule x3 Drawdown	2010	+29.3%	+30.3%	0.92	-22.0%	26

STRATEGY	YEAR	TOTAL RETURN	CAGR	SHARPE	MAX DD	TRADES
Sleeve @ 6%						
SPY Buy And Hold	2010	+15.8%	+16.3%	1.00	-12.3%	49
SPY Buy And Hold x3 @ 6%	2010	+27.3%	+28.2%	0.75	-36.0%	49
Sector Reserve Cash Rule	2010	+20.6%	+21.3%	1.08	-12.2%	26
ML Probability Rotation	2011	+15.6%	+15.9%	0.96	-12.7%	46
ML Probability Rotation x3 @ 6%	2011	+26.9%	+27.4%	0.74	-37.3%	46
ML Quality- Weighted Rotation	2011	+20.1%	+20.5%	1.18	-12.8%	46
ML Quality- Weighted Rotation x3 @ 6%	2011	+41.8%	+42.6%	0.96	-36.2%	46

STRATEGY	YEAR	TOTAL RETURN	CAGR	SHARPE	MAX DD	TRADES
Reserve Cash Rule x3 Drawdown Sleeve @ 6%	2011	+19.8%	+20.1%	0.72	-26.3%	27
SPY Buy And Hold	2011	-6.9%	-7.0%	-0.25	-19.7%	46
SPY Buy And Hold x3 @ 6%	2011	-37.3%	-37.8%	-0.43	-56.5%	46
Sector Reserve Cash Rule	2011	+17.1%	+17.4%	0.98	-14.1%	27
ML Probability Rotation	2012	+14.4%	+14.8%	2.12	-3.5%	35
ML Probability Rotation x3 @ 6%	2012	+33.7%	+34.7%	1.61	-10.7%	35
ML Quality- Weighted Rotation	2012	+13.7%	+14.1%	2.06	-3.5%	35
ML Quality- Weighted	2012	+31.4%	+32.3%	1.54	-10.6%	35

STRATEGY	YEAR	TOTAL RETURN	CAGR	SHARPE	MAX DD	TRADES
Rotation x3 @ 6%						
Reserve Cash Rule x3 Drawdown Sleeve @ 6%	2012	+10.2%	+10.5%	0.80	-11.6%	13
SPY Buy And Hold	2012	+7.2%	+7.4%	0.73	-7.9%	35
SPY Buy And Hold x3 @ 6%	2012	+8.0%	+8.2%	0.41	-27.5%	35
Sector Reserve Cash Rule	2012	+10.9%	+11.2%	1.71	-3.3%	13
ML Probability Rotation	2013	+25.9%	+26.5%	2.78	-3.4%	40
ML Probability Rotation x3 @ 6%	2013	+74.7%	+77.0%	2.35	-11.5%	40
ML Quality- Weighted Rotation	2013	+21.9%	+22.5%	2.41	-3.6%	40

STRATEGY	YEAR	TOTAL RETURN	CAGR	SHARPE	MAX DD	TRADES
ML Quality- Weighted Rotation x3 @ 6%	2013	+58.9%	+60.6%	1.97	-11.8%	40
Reserve Cash Rule x3 Drawdown Sleeve @ 6%	2013	+16.1%	+16.5%	2.93	-1.9%	1
SPY Buy And Hold	2013	+28.2%	+28.9%	3.00	-3.8%	40
SPY Buy And Hold x3 @ 6%	2013	+84.5%	+87.1%	2.57	-12.1%	40
Sector Reserve Cash Rule	2013	+15.7%	+16.1%	2.93	-1.9%	1
ML Probability Rotation	2014	+6.5%	+6.8%	0.59	-8.8%	41
ML Probability Rotation x3 @ 6%	2014	+4.4%	+4.6%	0.31	-27.7%	41
ML Quality-	2014	+7.5%	+7.8%	0.67	-8.9%	41

STRATEGY	YEAR	TOTAL RETURN	CAGR	SHARPE	MAX DD	TRADES
Weighted Rotation						
ML Quality- Weighted Rotation x3 @ 6%	2014	+7.5%	+7.8%	0.39	-26.5%	41
Reserve Cash Rule x3 Drawdown Sleeve @ 6%	2014	+8.4%	+8.8%	1.11	-5.2%	6
SPY Buy And Hold	2014	+11.6%	+12.0%	1.17	-6.0%	41
SPY Buy And Hold x3 @ 6%	2014	+22.0%	+22.9%	0.83	-18.5%	41
Sector Reserve Cash Rule	2014	+7.3%	+7.5%	1.01	-5.2%	6
ML Probability Rotation	2015	+15.2%	+16.7%	1.23	-6.6%	26
ML Probability Rotation x3 @ 6%	2015	+33.2%	+36.7%	0.98	-20.0%	26

STRATEGY	YEAR	TOTAL RETURN	CAGR	SHARPE	MAX DD	TRADES
ML Quality- Weighted Rotation	2015	+15.6%	+17.1%	1.22	-7.1%	26
ML Quality- Weighted Rotation x3 @ 6%	2015	+34.1%	+37.7%	0.97	-21.2%	26
Reserve Cash Rule x3 Drawdown Sleeve @ 6%	2015	+16.1%	+17.7%	1.49	-5.3%	6
SPY Buy And Hold	2015	-3.2%	-3.5%	-0.14	-13.9%	26
SPY Buy And Hold x3 @ 6%	2015	-23.8%	-25.7%	-0.36	-42.3%	26
Sector Reserve Cash Rule	2015	+13.4%	+14.7%	1.53	-4.5%	6
ML Probability Rotation	2016	+24.2%	+28.4%	2.03	-5.0%	28
ML Probability	2016	+64.9%	+78.3%	1.71	-15.4%	28

STRATEGY	YEAR	TOTAL RETURN	CAGR	SHARPE	MAX DD	TRADES
Rotation x3 @ 6%						
ML Quality- Weighted Rotation	2016	+23.0%	+27.0%	1.95	-5.0%	28
ML Quality- Weighted Rotation x3 @ 6%	2016	+60.3%	+72.5%	1.63	-15.4%	28
Reserve Cash Rule x3 Drawdown Sleeve @ 6%	2016	+20.7%	+24.2%	2.31	-3.0%	7
SPY Buy And Hold	2016	+20.3%	+23.9%	1.77	-4.0%	28
SPY Buy And Hold x3 @ 6%	2016	+50.4%	+60.2%	1.44	-13.9%	28
Sector Reserve Cash Rule	2016	+18.2%	+21.3%	2.39	-3.0%	7
ML Probability Rotation	2017	+7.3%	+7.5%	1.21	-3.3%	39

STRATEGY	YEAR	TOTAL RETURN	CAGR	SHARPE	MAX DD	TRADES
ML Probability Rotation x3 @ 6%	2017	+8.3%	+8.5%	0.54	-11.6%	39
ML Quality- Weighted Rotation	2017	+15.3%	+15.6%	2.18	-1.6%	39
ML Quality- Weighted Rotation x3 @ 6%	2017	+34.0%	+34.9%	1.58	-5.6%	39
Reserve Cash Rule x3 Drawdown Sleeve @ 6%	2017	+12.0%	+12.3%	2.90	-1.0%	0
SPY Buy And Hold	2017	+20.8%	+21.4%	3.26	-2.1%	39
SPY Buy And Hold x3 @ 6%	2017	+54.7%	+56.3%	2.57	-7.3%	39
Sector Reserve Cash Rule	2017	+12.0%	+12.3%	2.90	-1.0%	0

STRATEGY	YEAR	TOTAL RETURN	CAGR	SHARPE	MAX DD	TRADES
ML Probability Rotation	2018	+1.0%	+1.2%	0.16	-8.4%	23
ML Probability Rotation x3 @ 6%	2018	-10.8%	-12.9%	-0.18	-26.0%	23
ML Quality- Weighted Rotation	2018	-0.0%	-0.0%	0.06	-8.4%	23
ML Quality- Weighted Rotation x3 @ 6%	2018	-13.4%	-15.9%	-0.29	-26.8%	23
Reserve Cash Rule x3 Drawdown Sleeve @ 6%	2018	+4.6%	+5.6%	0.75	-4.9%	8
SPY Buy And Hold	2018	+3.7%	+4.5%	0.34	-9.1%	23
SPY Buy And Hold x3 @ 6%	2018	-7.8%	-9.3%	0.11	-29.7%	23

STRATEGY	YEAR	TOTAL RETURN	CAGR	SHARPE	MAX DD	TRADES
Sector Reserve Cash Rule	2018	+3.8%	+4.6%	0.65	-4.9%	8
ML Probability Rotation	2019	+14.4%	+26.5%	1.94	-5.9%	22
ML Probability Rotation x3 @ 6%	2019	+35.8%	+70.8%	1.61	-17.8%	22
ML Quality- Weighted Rotation	2019	+11.9%	+21.6%	1.55	-7.6%	22
ML Quality- Weighted Rotation x3 @ 6%	2019	+26.7%	+51.2%	1.23	-22.3%	22
Reserve Cash Rule x3 Drawdown Sleeve @ 6%	2019	+10.4%	+18.9%	1.81	-5.6%	12
SPY Buy And Hold	2019	+8.0%	+14.3%	1.07	-6.6%	22

STRATEGY	YEAR	TOTAL RETURN	CAGR	SHARPE	MAX DD	TRADES
SPY Buy And Hold x3 @ 6%	2019	+13.6%	+25.0%	0.76	-19.8%	22
Sector Reserve Cash Rule	2019	+9.4%	+17.1%	1.89	-4.8%	12
ML Probability Rotation	2020	+14.6%	+21.9%	1.39	-6.7%	19
ML Probability Rotation x3 @ 6%	2020	+31.7%	+49.3%	1.12	-21.8%	19
ML Quality- Weighted Rotation	2020	+17.3%	+26.2%	1.54	-5.8%	19
ML Quality- Weighted Rotation x3 @ 6%	2020	+40.6%	+64.3%	1.28	-19.7%	19
Reserve Cash Rule x3 Drawdown Sleeve @ 6%	2020	+12.4%	+18.6%	1.47	-4.2%	11

STRATEGY	YEAR	TOTAL RETURN	CAGR	SHARPE	MAX DD	TRADES
SPY Buy And Hold	2020	+34.3%	+53.6%	2.35	-8.1%	19
SPY Buy And Hold x3 @ 6%	2020	+105.4%	+185.1%	2.14	-23.9%	19
Sector Reserve Cash Rule	2020	+12.4%	+18.5%	1.69	-3.6%	11
ML Probability Rotation	2021	+33.3%	+34.2%	2.62	-4.6%	29
ML Probability Rotation x3 @ 6%	2021	+101.9%	+105.2%	2.26	-14.9%	29
ML Quality- Weighted Rotation	2021	+28.9%	+29.7%	2.31	-5.2%	29
ML Quality- Weighted Rotation x3 @ 6%	2021	+82.8%	+85.4%	1.96	-15.8%	29
Reserve Cash Rule x3 Drawdown	2021	+21.0%	+21.5%	2.87	-3.1%	0

STRATEGY	YEAR	TOTAL RETURN	CAGR	SHARPE	MAX DD	TRADES
Sleeve @ 6%						
SPY Buy And Hold	2021	+31.2%	+32.0%	2.46	-4.7%	29
SPY Buy And Hold x3 @ 6%	2021	+92.2%	+95.1%	2.11	-14.6%	29
Sector Reserve Cash Rule	2021	+21.0%	+21.5%	2.87	-3.1%	0
ML Probability Rotation	2022	+1.3%	+1.4%	0.17	-18.2%	44
ML Probability Rotation x3 @ 6%	2022	-20.5%	-20.8%	-0.01	-55.5%	44
ML Quality- Weighted Rotation	2022	+1.7%	+1.7%	0.19	-18.8%	44
ML Quality- Weighted Rotation x3 @ 6%	2022	-21.0%	-21.4%	0.02	-57.2%	44

STRATEGY	YEAR	TOTAL RETURN	CAGR	SHARPE	MAX DD	TRADES
Reserve Cash Rule x3 Drawdown Sleeve @ 6%	2022	+10.7%	+10.9%	0.47	-32.2%	23
SPY Buy And Hold	2022	-19.8%	-20.2%	-0.83	-25.3%	44
SPY Buy And Hold x3 @ 6%	2022	-61.9%	-62.6%	-1.00	-67.3%	44
Sector Reserve Cash Rule	2022	+7.2%	+7.4%	0.44	-17.8%	23
ML Probability Rotation	2023	-0.2%	-0.2%	0.05	-9.2%	34
ML Probability Rotation x3 @ 6%	2023	-14.7%	-15.0%	-0.26	-31.7%	34
ML Quality- Weighted Rotation	2023	+0.4%	+0.5%	0.10	-9.3%	34
ML Quality- Weighted	2023	-12.9%	-13.2%	-0.21	-31.3%	34

STRATEGY	YEAR	TOTAL RETURN	CAGR	SHARPE	MAX DD	TRADES
Rotation x3 @ 6%						
Reserve Cash Rule x3 Drawdown Sleeve @ 6%	2023	-18.4%	-18.8%	-0.63	-27.4%	18
SPY Buy And Hold	2023	+19.9%	+20.5%	1.48	-7.6%	34
SPY Buy And Hold x3 @ 6%	2023	+47.1%	+48.5%	1.20	-24.0%	34
Sector Reserve Cash Rule	2023	-4.0%	-4.1%	-0.25	-12.0%	18
ML Probability Rotation	2024	+18.3%	+18.7%	2.40	-1.2%	16
ML Probability Rotation x3 @ 6%	2024	+47.6%	+48.6%	1.93	-6.1%	16
ML Quality- Weighted Rotation	2024	+17.3%	+17.6%	2.46	-1.2%	16

STRATEGY	YEAR	TOTAL RETURN	CAGR	SHARPE	MAX DD	TRADES
ML Quality- Weighted Rotation x3 @ 6%	2024	+44.2%	+45.1%	1.95	-6.0%	16
Reserve Cash Rule x3 Drawdown Sleeve @ 6%	2024	+20.4%	+20.8%	2.99	-0.8%	5
SPY Buy And Hold	2024	+15.7%	+16.0%	1.49	-4.9%	16
SPY Buy And Hold x3 @ 6%	2024	+35.7%	+36.4%	1.16	-16.7%	16
Sector Reserve Cash Rule	2024	+16.2%	+16.5%	3.38	-0.5%	5
ML Probability Rotation	2025	+0.6%	+0.7%	0.12	-3.7%	17
ML Probability Rotation x3 @ 6%	2025	-4.8%	-4.9%	-0.12	-11.5%	17
ML Quality-	2025	+2.9%	+3.0%	0.38	-3.8%	17

STRATEGY	YEAR	TOTAL RETURN	CAGR	SHARPE	MAX DD	TRADES
Weighted Rotation						
ML Quality- Weighted Rotation x3 @ 6%	2025	+1.2%	+1.2%	0.17	-12.0%	17
Reserve Cash Rule x3 Drawdown Sleeve @ 6%	2025	+3.4%	+3.4%	0.59	-2.4%	7
SPY Buy And Hold	2025	+14.7%	+15.1%	1.78	-2.5%	17
SPY Buy And Hold x3 @ 6%	2025	+40.5%	+41.6%	1.56	-7.8%	17
Sector Reserve Cash Rule	2025	+3.1%	+3.2%	0.61	-2.2%	7
ML Probability Rotation	2026	+8.6%	n/a	n/a	-1.1%	8
ML Probability Rotation x3 @ 6%	2026	+24.4%	n/a	n/a	-3.4%	8

STRATEGY	YEAR	TOTAL RETURN	CAGR	SHARPE	MAX DD	TRADES
ML Quality- Weighted Rotation	2026	+7.8%	n/a	n/a	-1.2%	8
ML Quality- Weighted Rotation x3 @ 6%	2026	+21.5%	n/a	n/a	-3.8%	8
Reserve Cash Rule x3 Drawdown Sleeve @ 6%	2026	+5.0%	n/a	n/a	-0.7%	0
SPY Buy And Hold	2026	-1.0%	n/a	n/a	-2.0%	8
SPY Buy And Hold x3 @ 6%	2026	-5.2%	n/a	n/a	-7.2%	8
Sector Reserve Cash Rule	2026	+5.0%	n/a	n/a	-0.7%	0

STRATEGY	SIGNAL REGIME	TOTAL RETURN	CAGR	SHARPE	MAX DD	HIT RATE
ML Probability Rotation	Cash	-13.4%	-0.7%	-2.94	-13.4%	0%
ML Probability Rotation x3 @ 6%	Cash	-62.0%	-4.8%	-6.45	-61.9%	0%
ML Quality- Weighted Rotation	Cash	-13.4%	-0.7%	-2.94	-13.4%	0%
ML Quality- Weighted Rotation x3 @ 6%	Cash	-62.0%	-4.8%	-6.45	-61.9%	0%
Reserve Cash Rule x3 Drawdown Sleeve @ 6%	Cash	-19.4%	-1.1%	-0.24	-28.2%	77%
SPY Buy And Hold	Cash	-51.2%	-3.6%	-0.52	-55.9%	49%
SPY Buy And Hold x3 @ 6%	Cash	-94.9%	-14.2%	-0.66	-96.0%	45%

STRATEGY	SIGNAL REGIME	TOTAL RETURN	CAGR	SHARPE	MAX DD	HIT RA
Sector Reserve Cash Rule	Cash	+2.2%	+0.1%	0.08	-6.1%	81%
ML Probability Rotation	Credit Deleveraging	+35.3%	+2.5%	0.50	-9.9%	65%
ML Probability Rotation x3 @ 6%	Credit Deleveraging	+104.7%	+6.0%	0.46	-29.5%	65%
ML Quality- Weighted Rotation	Credit Deleveraging	+33.1%	+2.3%	0.43	-11.9%	65%
ML Quality- Weighted Rotation x3 @ 6%	Credit Deleveraging	+89.8%	+5.3%	0.39	-34.8%	65%
Reserve Cash Rule x3 Drawdown Sleeve @ 6%	Credit Deleveraging	+57.2%	+3.7%	0.42	-20.8%	65%
SPY Buy And Hold	Credit Deleveraging	+39.7%	+2.8%	0.53	-10.2%	68%

STRATEGY	SIGNAL REGIME	TOTAL RETURN	CAGR	SHARPE	MAX DD	HIT RA
SPY Buy And Hold x3 @ 6%	Credit Deleveraging	+123.2%	+6.7%	0.49	-29.8%	68%
Sector Reserve Cash Rule	Credit Deleveraging	+33.1%	+2.3%	0.45	-11.7%	65%
ML Probability Rotation	Disinflationary Growth	+9.7%	+0.7%	0.15	-19.4%	55%
ML Probability Rotation x3 @ 6%	Disinflationary Growth	-17.4%	-1.5%	0.02	-55.4%	53%
ML Quality- Weighted Rotation	Disinflationary Growth	+11.9%	+0.9%	0.17	-21.0%	57%
ML Quality- Weighted Rotation x3 @ 6%	Disinflationary Growth	-13.2%	-1.1%	0.05	-58.1%	55%
Reserve Cash Rule x3 Drawdown Sleeve @ 6%	Disinflationary Growth	+0.1%	+0.0%	0.03	-24.3%	57%

STRATEGY	SIGNAL REGIME	TOTAL RETURN	CAGR	SHARPE	MAX DD	HIT RAT
SPY Buy And Hold	Disinflationary Growth	+25.0%	+1.8%	0.32	-17.2%	58%
SPY Buy And Hold x3 @ 6%	Disinflationary Growth	+22.8%	+1.7%	0.19	-50.8%	55%
Sector Reserve Cash Rule	Disinflationary Growth	+9.0%	+0.7%	0.17	-15.7%	57%
ML Probability Rotation	Fragile Late- Cycle Watch	+322.3%	+7.9%	0.98	-14.7%	66%
ML Probability Rotation x3 @ 6%	Fragile Late- Cycle Watch	+2663.5%	+19.1%	0.85	-45.3%	65%
ML Quality- Weighted Rotation	Fragile Late- Cycle Watch	+347.0%	+8.2%	0.96	-14.1%	67%
ML Quality- Weighted Rotation x3 @ 6%	Fragile Late- Cycle Watch	+3010.4%	+19.8%	0.83	-45.2%	65%
Reserve Cash Rule x3 Drawdown	Fragile Late- Cycle Watch	+347.3%	+8.2%	0.75	-24.7%	65%

STRATEGY	SIGNAL REGIME	TOTAL RETURN	CAGR	SHARPE	MAX DD	HIT RAT
Sleeve @ 6%						
SPY Buy And Hold	Fragile Late- Cycle Watch	+228.5%	+6.4%	0.81	-21.2%	65%
SPY Buy And Hold x3 @ 6%	Fragile Late- Cycle Watch	+1176.2%	+14.3%	0.68	-58.0%	63%
Sector Reserve Cash Rule	Fragile Late- Cycle Watch	+230.3%	+6.5%	0.93	-13.4%	67%
ML Probability Rotation	Inflationary Boom	+26.2%	+1.3%	0.26	-17.6%	58%
ML Probability Rotation x3 @ 6%	Inflationary Boom	+34.7%	+1.7%	0.19	-50.4%	56%
ML Quality- Weighted Rotation	Inflationary Boom	+21.5%	+1.1%	0.21	-20.3%	57%
ML Quality- Weighted Rotation x3 @ 6%	Inflationary Boom	+18.1%	+0.9%	0.14	-55.8%	54%

STRATEGY	SIGNAL REGIME	TOTAL RETURN	CAGR	SHARPE	MAX DD	HIT RATE
Reserve Cash Rule x3 Drawdown Sleeve @ 6%	Inflationary Boom	+9.5%	+0.5%	0.10	-28.7%	56%
SPY Buy And Hold	Inflationary Boom	+55.3%	+2.5%	0.51	-14.1%	64%
SPY Buy And Hold x3 @ 6%	Inflationary Boom	+160.5%	+5.6%	0.43	-43.0%	62%
Sector Reserve Cash Rule	Inflationary Boom	+16.0%	+0.8%	0.18	-17.6%	58%
ML Probability Rotation	Liquidity Bubble Or Valuation Stretch	+28.7%	+5.3%	1.23	-3.3%	75%
ML Probability Rotation x3 @ 6%	Liquidity Bubble Or Valuation Stretch	+93.4%	+14.5%	1.12	-11.5%	71%
ML Quality-Weighted Rotation	Liquidity Bubble Or Valuation Stretch	+28.1%	+5.2%	1.21	-3.4%	71%
ML Quality-	Liquidity Bubble Or	+90.6%	+14.1%	1.10	-11.7%	68%

STRATEGY	SIGNAL REGIME	TOTAL RETURN	CAGR	SHARPE	MAX DD	HIT RATE
Weighted Rotation x3 @ 6%	Valuation Stretch					
Reserve Cash Rule x3 Drawdown Sleeve @ 6%	Liquidity Bubble Or Valuation Stretch	+19.2%	+3.7%	1.25	-2.1%	75%
SPY Buy And Hold	Liquidity Bubble Or Valuation Stretch	+19.2%	+3.7%	1.04	-2.2%	71%
SPY Buy And Hold x3 @ 6%	Liquidity Bubble Or Valuation Stretch	+55.2%	+9.4%	0.91	-7.2%	64%
Sector Reserve Cash Rule	Liquidity Bubble Or Valuation Stretch	+18.0%	+3.5%	1.28	-1.9%	75%
ML Probability Rotation	Panic Or Forced Liquidation	-28.9%	n/a	n/a	-21.2%	32%
ML Probability Rotation x3 @ 6%	Panic Or Forced Liquidation	-71.4%	n/a	n/a	-58.0%	32%

STRATEGY	SIGNAL REGIME	TOTAL RETURN	CAGR	SHARPE	MAX DD	HIT RATE
ML Quality-Weighted Rotation	Panic Or Forced Liquidation	-29.9%	n/a	n/a	-22.2%	36%
ML Quality-Weighted Rotation x3 @ 6%	Panic Or Forced Liquidation	-72.7%	n/a	n/a	-59.7%	36%
Reserve Cash Rule x3 Drawdown Sleeve @ 6%	Panic Or Forced Liquidation	-42.1%	n/a	n/a	-38.0%	41%
SPY Buy And Hold	Panic Or Forced Liquidation	-36.9%	n/a	n/a	-25.8%	41%
SPY Buy And Hold x3 @ 6%	Panic Or Forced Liquidation	-82.9%	n/a	n/a	-68.1%	41%
Sector Reserve Cash Rule	Panic Or Forced Liquidation	-26.2%	n/a	n/a	-21.9%	41%
ML Probability Rotation	Rate-Shock Regime	+0.2%	n/a	n/a	-14.3%	57%

STRATEGY	SIGNAL REGIME	TOTAL RETURN	CAGR	SHARPE	MAX DD	HIT RA
ML Probability Rotation x3 @ 6%	Rate-Shock Regime	-6.5%	n/a	n/a	-38.9%	57%
ML Quality- Weighted Rotation	Rate-Shock Regime	+2.8%	n/a	n/a	-12.4%	57%
ML Quality- Weighted Rotation x3 @ 6%	Rate-Shock Regime	+1.0%	n/a	n/a	-34.6%	57%
Reserve Cash Rule x3 Drawdown Sleeve @ 6%	Rate-Shock Regime	+16.6%	n/a	n/a	-11.1%	57%
SPY Buy And Hold	Rate-Shock Regime	+3.0%	n/a	n/a	-11.1%	57%
SPY Buy And Hold x3 @ 6%	Rate-Shock Regime	+2.1%	n/a	n/a	-31.5%	57%
Sector Reserve Cash Rule	Rate-Shock Regime	+7.1%	n/a	n/a	-8.7%	57%

STRATEGY	SIGNAL REGIME	TOTAL RETURN	CAGR	SHARPE	MAX DD	HIT RA
ML Probability Rotation	Recovery And Reflation	+18.0%	+6.7%	1.26	-2.4%	64%
ML Probability Rotation x3 @ 6%	Recovery And Reflation	+52.2%	+17.9%	1.13	-7.3%	64%
ML Quality- Weighted Rotation	Recovery And Reflation	+19.6%	+7.3%	1.12	-3.4%	68%
ML Quality- Weighted Rotation x3 @ 6%	Recovery And Reflation	+56.8%	+19.3%	1.01	-10.5%	68%
Reserve Cash Rule x3 Drawdown Sleeve @ 6%	Recovery And Reflation	+30.0%	+10.9%	0.85	-7.3%	68%
SPY Buy And Hold	Recovery And Reflation	+21.1%	+7.8%	1.37	-2.9%	68%
SPY Buy And Hold x3 @ 6%	Recovery And Reflation	+64.1%	+21.4%	1.25	-8.8%	68%

STRATEGY	SIGNAL REGIME	TOTAL RETURN	CAGR	SHARPE	MAX DD	HIT RATE
Sector Reserve Cash Rule	Recovery And Reflation	+18.5%	+6.9%	0.99	-3.7%	68%
ML Probability Rotation	Sideways Low-Volatility Regime	+11.0%	+1.4%	0.37	-6.1%	61%
ML Probability Rotation x3 @ 6%	Sideways Low-Volatility Regime	+23.1%	+2.9%	0.29	-18.5%	61%
ML Quality-Weighted Rotation	Sideways Low-Volatility Regime	+9.6%	+1.3%	0.32	-7.7%	61%
ML Quality-Weighted Rotation x3 @ 6%	Sideways Low-Volatility Regime	+18.5%	+2.4%	0.25	-22.8%	61%
Reserve Cash Rule x3 Drawdown Sleeve @ 6%	Sideways Low-Volatility Regime	+9.8%	+1.3%	0.40	-5.6%	61%
SPY Buy And Hold	Sideways Low-Volatility Regime	+14.1%	+1.8%	0.55	-6.6%	61%

STRATEGY	SIGNAL REGIME	TOTAL RETURN	CAGR	SHARPE	MAX DD	HIT RATE
SPY Buy And Hold x3 @ 6%	Sideways Low- Volatility Regime	+35.4%	+4.3%	0.46	-19.8%	57%
Sector Reserve Cash Rule	Sideways Low- Volatility Regime	+8.0%	+1.1%	0.39	-4.8%	61%
ML Probability Rotation	Unknown	+1.1%	n/a	n/a	-4.7%	56%
ML Probability Rotation x3 @ 6%	Unknown	-1.1%	n/a	n/a	-14.5%	44%
ML Quality- Weighted Rotation	Unknown	+0.8%	n/a	n/a	-4.9%	50%
ML Quality- Weighted Rotation x3 @ 6%	Unknown	-2.1%	n/a	n/a	-15.5%	38%
Reserve Cash Rule x3 Drawdown Sleeve @ 6%	Unknown	+1.4%	n/a	n/a	-3.3%	50%

STRATEGY	SIGNAL REGIME	TOTAL RETURN	CAGR	SHARPE	MAX DD	HIT RAT
SPY Buy And Hold	Unknown	+2.8%	n/a	n/a	-1.5%	31%
SPY Buy And Hold x3 @ 6%	Unknown	+3.9%	n/a	n/a	-5.4%	31%
Sector Reserve Cash Rule	Unknown	+1.3%	n/a	n/a	-2.9%	50%

STRESS WINDOWS

Deep Drawdown Windows And Reserve Triggers

This table isolates the worst SPY drawdown signals inside the full 2006-2026 walk-forward history so you can see how the reserve rule behaved in the key stress episodes.

SIGNAL DATE	ENTRY DATE	EXIT DATE	SPY DRAWDOWN	REGIME	QUALITY SELECTION	RESERVE SECTOR
2009-03-03	2009-03-04	2009-03-11	-53.9%	Panic Or Forced Liquidation	XLB, XLK, XLY	XLB
2009-02-24	2009-02-25	2009-03-04	-49.0%	Panic Or Forced Liquidation	XLB, XLI, XLK	XLB
2009-02-17	2009-02-18	2009-02-25	-47.9%	Panic Or Forced Liquidation	XLB, XLI	XLB

SIGNAL DATE	ENTRY DATE	EXIT DATE	SPY DRAWDOWN	REGIME	QUALITY SELECTION	RESI SECT
2009-02-02	2009-02-03	2009-02-10	-45.7%	Panic Or Forced Liquidation	XLB, XLI, XLK	XLB
2009-04-08	2009-04-09	2009-04-17	-45.3%	Credit Deleveraging	XLB, XLI, XLK	XLB
2009-01-26	2009-01-27	2009-02-03	-44.9%	Panic Or Forced Liquidation	XLI, XLB, XLY	XLI
2008-11-25	2008-11-26	2008-12-04	-44.1%	Panic Or Forced Liquidation	XLP	XLP
2009-01-16	2009-01-20	2009-01-27	-44.0%	Panic Or Forced Liquidation	XLB, XLI, XLK	XLB
2009-04-23	2009-04-24	2009-05-01	-43.4%	Credit Deleveraging	XLB, XLI, XLK	XLB
2008-11-18	2008-11-19	2008-11-26	-43.2%	Panic Or Forced Liquidation	XLP	XLP
2008-12-03	2008-12-04	2008-12-11	-43.0%	Panic Or Forced Liquidation	XLB, XLY, XLP	XLB
2008-12-24	2008-12-26	2009-01-05	-43.0%	Panic Or Forced Liquidation	XLB, XLK, XLP	XLB

SIGNAL DATE	ENTRY DATE	EXIT DATE	SPY DRAWDOWN	REGIME	QUALITY SELECTION	RESERV SECTOR
2009-02-09	2009-02-10	2009-02-18	-42.7%	Panic Or Forced Liquidation	XLB, XLY, XLP	XLB
2009-04-16	2009-04-17	2009-04-24	-42.7%	Cash	Cash	XLB
2009-04-30	2009-05-01	2009-05-08	-42.1%	Credit Deleveraging	XLY, XLV, XLP	XLY

REBALANCE LOG

Latest Holdout Rotation Decisions

The log below shows the most recent holdout rebalance windows, the sectors selected by the quality-weighted ML score, and the realized next-window return compared with SPY.

SIGNAL DATE	ENTRY DATE	EXIT DATE	SPY DRAWDOWN	REGIME	QUALITY SELECTION	RESERV SECTOR
2026-03-03	2026-03-04	2026-03-11	-2.2%	Liquidity Bubble Or Valuation Stretch	XLU	XLU
2026-02-24	2026-02-25	2026-03-04	-1.2%	Liquidity Bubble Or Valuation Stretch	XLU	XLU

SIGNAL DATE	ENTRY DATE	EXIT DATE	SPY DRAWDOWN	REGIME	QUALITY SELECTION	RESERV SECTOR
2026- 02-17	2026- 02-18	2026- 02-25	-1.8%	Liquidity Bubble Or Valuation Stretch	XLI, XLY, XLU	XLI
2026- 02-09	2026- 02-10	2026- 02-18	-0.2%	Liquidity Bubble Or Valuation Stretch	XLU	XLU
2026- 02-02	2026- 02-03	2026- 02-10	-0.0%	Liquidity Bubble Or Valuation Stretch	XLU	XLU
2026- 01-26	2026- 01-27	2026- 02-03	-0.3%	Liquidity Bubble Or Valuation Stretch	XLI, XLY, XLU	XLI
2026- 01-16	2026- 01-20	2026- 01-27	-0.5%	Liquidity Bubble Or Valuation Stretch	XLY	XLY
2026- 01-09	2026- 01-12	2026- 01-20	+0.0%	Liquidity Bubble Or	XLY, XLU	XLY

SIGNAL DATE	ENTRY DATE	EXIT DATE	SPY DRAWDOWN	REGIME	QUALITY SELECTION	RESERV SECTOR
				Valuation Stretch		
2026- 01-02	2026- 01-05	2026- 01-12	-1.0%	Liquidity Bubble Or Valuation Stretch	XLY, XLE, XLU	XLY
2025- 12-24	2025- 12-26	2026- 01-05	+0.0%	Liquidity Bubble Or Valuation Stretch	XLY, XLU	XLY
2025- 12-17	2025- 12-18	2025- 12-26	-2.6%	Liquidity Bubble Or Valuation Stretch	XLI, XLY, XLE	XLI
2025- 12-10	2025- 12-11	2025- 12-18	+0.0%	Liquidity Bubble Or Valuation Stretch	XLI, XLY, XLE	XLI

SECTOR MAP

The Equity Buckets Scored In This Report

XLE • REAL ASSET CYCLICAL

XLB • CYCLICAL INFLATION BETA

Energy

High earnings leverage to oil, capex, and nominal GDP.

Often benefits most when inflation and supply shocks dominate the tape.

Commodity / Materials

Sensitive to industrial demand, pricing power, and commodity volumes.

Usually improves when nominal growth and input scarcity rise together.

XLI • GROWTH AND CAPEX CYCLICAL

Manufacturing / Industrials

Leverages industrial production, capex, and freight intensity.

Tends to lead in recovery and reflation regimes when production broadens.

XLY • CONSUMER CYCLICAL

Retail / Consumer Discretionary

Most exposed to real income, labor resilience, and disinflation relief.

Usually wants falling input pressure, healthy labor, and calm credit.

XLK • DURATION AND INNOVATION

Information Technology

Benefits from disinflation, lower discount rates, and enterprise capex strength.

Typically strongest in disinflationary growth or liquidity-led melt-up regimes.

XLP • DEFENSIVE CASH FLOW

Consumer Staples

Lower-volatility earnings with less macro beta than cyclical retail.

Acts as a defensive buffer when growth is fading or rate stress rises.

XLV • DEFENSIVE QUALITY

Health Care

Relatively stable earnings stream with lower sensitivity to the goods cycle.

XLU • RATE-SENSITIVE DEFENSE

Utilities

Stable regulated cash flows, but still sensitive to rate and duration

Useful when the portfolio needs defense without fully abandoning equity exposure.

pressure.

Can cushion drawdowns when stress rises, though not every inflation regime helps it.

ALLOCATION

Current 40 / 60 Portfolio Construction

The allocation below distributes the 60% equity sleeve across the highest-scoring sector buckets under the current regime. The remaining 40% stays in cash by rule.

SECTOR	SLEEVE WEIGHT	PORTFOLIO WEIGHT	EXPECTED 1Y	EXPECTED 3Y	HIGHER HIGH 12M
Information Technology (XLK)	33.8%	20.3%	+16.7%	+15.1%	90%
Consumer Staples (XLP)	22.5%	13.5%	+11.4%	+10.5%	93%
Retail / Consumer Discretionary (XLY)	22.5%	13.5%	+14.9%	+14.0%	88%
Health Care (XLV)	17.4%	10.4%	+13.9%	+13.7%	85%
Manufacturing / Industrials (XLI)	3.8%	2.3%	+11.9%	+11.8%	81%

CURRENT REGIME MATRIX

All Sector Scores For The Active Regime

SECTOR	TYPE	EXPECTED 1Y	EXPECTED 3Y	HIGHER HIGH 12M	AVG DRAWDOW 12M
Information Technology (XLK)	Duration and innovation	+16.7%	+15.1%	90%	-4.5%
Consumer Staples (XLP)	Defensive cash flow	+11.4%	+10.5%	93%	-2.3%
Retail / Consumer Discretionary (XLY)	Consumer cyclical	+14.9%	+14.0%	88%	-5.0%
Health Care (XLV)	Defensive quality	+13.9%	+13.7%	85%	-3.6%
Manufacturing / Industrials (XLI)	Growth and capex cyclical	+11.9%	+11.8%	81%	-7.4%
Utilities (XLU)	Rate-sensitive defense	+9.2%	+8.7%	89%	-3.8%
Commodity / Materials (XLB)	Cyclical inflation beta	+10.1%	+8.7%	78%	-9.2%

SECTOR	TYPE	EXPECTED 1Y	EXPECTED 3Y	HIGHER HIGH 12M	AVG DRAWDOW 12M
Energy (XLE)	Real asset cyclical	+8.5%	+8.1%	55%	-12.6%

REGIME BEHAVIOUR

Where Drawdowns Clustered And Where Higher Highs Happened

WORST DRAWDOWN REGIME

Panic Or Forced Liquidation

Average sector drawdown -13.9%.
Higher-high rate 0%. Least-damaged sectors: Retail / Consumer Discretionary, Information Technology, Commodity / Materials.

WORST DRAWDOWN REGIME

Credit Deleveraging

Average sector drawdown -13.6%.
Higher-high rate 39%. Least-damaged sectors: Retail / Consumer Discretionary, Manufacturing / Industrials, Information Technology.

WORST DRAWDOWN REGIME

Fragile Late-Cycle Watch

Average sector drawdown -6.0%.
Higher-high rate 82%. Least-damaged sectors: Information Technology, Retail / Consumer Discretionary, Health Care.

HIGHER-HIGH REGIME

Sideways Low-Volatility Regime

Higher-high hit rate 95%. Average drawdown -5.7%. Most frequent leaders: Information Technology, Retail / Consumer Discretionary, Health Care.

HIGHER-HIGH REGIME

HIGHER-HIGH REGIME

Disinflationary Growth

Liquidity Bubble Or Valuation Stretch

Higher-high hit rate 94%. Average drawdown -3.1%. Most frequent leaders: Information Technology, Utilities, Energy.

Higher-high hit rate 88%. Average drawdown -4.8%. Most frequent leaders: Information Technology, Utilities, Consumer Staples.

REGIME	AVG 1Y SECTOR RETURN	AVG 3Y SECTOR RETURN	HIGHER HIGH 12M	AVG DRAWDOWN 12M	AVG CONFIDENCE
Panic Or Forced Liquidation	+24.2%	+16.0%	0%	-13.9%	49
Recovery And Reflation	+13.8%	+14.4%	77%	-1.0%	69
Sideways Low- Volatility Regime	+13.4%	+12.0%	95%	-5.7%	85
Fragile Late- Cycle Watch	+12.1%	+11.3%	82%	-6.0%	89

REGIME	AVG 1Y SECTOR RETURN	AVG 3Y SECTOR RETURN	HIGHER HIGH 12M	AVG DRAWDOWN 12M	AVG CONFIDENCE
Liquidity Bubble Or Valuation Stretch	+11.8%	+9.9%	94%	-3.1%	74
Inflationary Boom	+11.8%	+9.1%	75%	-4.9%	88
Credit Deleveraging	+9.0%	+12.2%	39%	-13.6%	54
Disinflationary Growth	+8.5%	+7.6%	88%	-4.8%	88

SECTOR ML

Per-Sector Ensemble Results

The table below uses the simple averaged ensemble as the live signal and separately identifies which single model preserved validation behavior into the untouched holdout best.

SECTOR	BEST ANTI-OVERFIT MODEL	STABILITY SCORE	HOLDOUT CAGR	HOLDOUT SHARPE	HOLDOUT SORTINO
Information Technology (XLK)	LightGBM	99.6	+6.7%	1.71	n/a
Utilities (XLU)	ExtraTrees	98.8	+16.7%	2.87	3.65
Manufacturing / Industrials (XLI)	Average Ensemble	98.6	+11.2%	2.24	n/a
Energy (XLE)	Average Ensemble	97.7	+3.1%	0.72	0.20
Retail / Consumer Discretionary (XLY)	Average Ensemble	96.1	+5.0%	0.78	0.62
Commodity / Materials (XLB)	ExtraTrees	94.6	+2.4%	0.74	n/a
Consumer Staples (XLP)	LightGBM	93.8	+0.2%	0.12	0.06
Health Care (XLV)	ExtraTrees	78.7	-1.5%	-1.32	n/a

MODEL AUDIT

Validation-To-Holdout Stability By Model

Higher stability scores mean the model degraded less from walk-forward validation into the untouched holdout. This is the direct overfitting check, not just a performance ranking.

SECTOR	MODEL	STABILITY	VAL AUC	HOLDOUT AUC	VAL BRIER	HOLD BRIER
Commodity / Materials (XLB)	ExtraTrees	94.6	0.558	0.649	0.248	0.257
Commodity / Materials (XLB)	Elastic Net	94.3	0.558	0.573	0.303	0.302
Commodity / Materials (XLB)	LightGBM	93.8	0.597	0.663	0.249	0.263
Commodity / Materials (XLB)	Average Ensemble	87.7	0.593	0.644	0.246	0.270
Consumer Staples (XLP)	LightGBM	93.8	0.558	0.516	0.257	0.262
Consumer Staples (XLP)	Elastic Net	84.3	0.520	0.432	0.304	0.295
Consumer Staples (XLP)	ExtraTrees	83.1	0.542	0.441	0.249	0.257
Consumer Staples (XLP)	Average Ensemble	81.3	0.542	0.450	0.255	0.267

SECTOR	MODEL	STABILITY	VAL AUC	HOLDOUT AUC	VAL BRIER	HOLD BRIER
Energy (XLE)	Average Ensemble	97.7	0.545	0.572	0.259	0.253
Energy (XLE)	Elastic Net	94.0	0.533	0.572	0.329	0.235
Energy (XLE)	LightGBM	92.5	0.536	0.520	0.267	0.283
Energy (XLE)	ExtraTrees	83.8	0.546	0.567	0.248	0.262
Health Care (XLV)	ExtraTrees	78.7	0.529	0.411	0.249	0.265
Health Care (XLV)	LightGBM	69.5	0.555	0.562	0.255	0.275
Health Care (XLV)	Average Ensemble	63.9	0.555	0.599	0.250	0.284
Health Care (XLV)	Elastic Net	60.3	0.540	0.571	0.291	0.338
Information Technology (XLK)	LightGBM	99.6	0.578	0.581	0.253	0.253
Information Technology (XLK)	ExtraTrees	96.0	0.531	0.615	0.248	0.257
Information Technology (XLK)	Average Ensemble	88.0	0.583	0.574	0.249	0.280

SECTOR	MODEL	STABILITY	VAL AUC	HOLDOUT AUC	VAL BRIER	HOLD BRIER
Information Technology (XLK)	Elastic Net	74.9	0.562	0.518	0.303	0.374
Manufacturing / Industrials (XLI)	Average Ensemble	98.6	0.587	0.615	0.246	0.245
Manufacturing / Industrials (XLI)	LightGBM	96.1	0.584	0.623	0.251	0.259
Manufacturing / Industrials (XLI)	ExtraTrees	94.2	0.540	0.640	0.248	0.252
Manufacturing / Industrials (XLI)	Elastic Net	91.6	0.563	0.538	0.299	0.243
Retail / Consumer Discretionary (XLY)	Average Ensemble	96.1	0.563	0.554	0.252	0.245
Retail / Consumer Discretionary (XLY)	Elastic Net	94.7	0.535	0.514	0.308	0.264
Retail / Consumer Discretionary (XLY)	LightGBM	94.6	0.586	0.602	0.252	0.245

SECTOR	MODEL	STABILITY	VAL AUC	HOLDOUT AUC	VAL BRIER	HOLD BRIER
Retail / Consumer Discretionary (XLY)	ExtraTrees	93.9	0.546	0.554	0.248	0.249
Utilities (XLU)	ExtraTrees	98.8	0.523	0.528	0.250	0.249
Utilities (XLU)	LightGBM	97.2	0.531	0.562	0.266	0.246
Utilities (XLU)	Average Ensemble	85.2	0.529	0.541	0.261	0.237
Utilities (XLU)	Elastic Net	82.5	0.522	0.508	0.311	0.255

SENSITIVITY

Fee And Slippage Sensitivity On The Holdout

The model is trained on a 15 bps all-in cost assumption. The table below replays the untouched holdout under separate fee and slippage stress scenarios.

SECTOR	STRESS TYPE	SCENARIO	TOTAL COST	HOLDOUT CAGR	HOLDOUT SHARPE	H M
Commodity / Materials (XLB)	Fee	5 bps	10 bps	+2.5%	0.76	-1
Commodity / Materials	Fee	10 bps	15 bps	+2.4%	0.74	-1

SECTOR	STRESS TYPE	SCENARIO	TOTAL COST	HOLDOUT CAGR	HOLDOUT SHARPE	H M
(XLB)						
Commodity / Materials (XLB)	Fee	20 bps	25 bps	+2.3%	0.69	-2
Commodity / Materials (XLB)	Slippage	0 bps	10 bps	+2.5%	0.76	-1
Commodity / Materials (XLB)	Slippage	5 bps	15 bps	+2.4%	0.74	-1
Commodity / Materials (XLB)	Slippage	10 bps	20 bps	+2.4%	0.71	-2
Consumer Staples (XLP)	Fee	5 bps	10 bps	+0.3%	0.23	-1
Consumer Staples (XLP)	Fee	10 bps	15 bps	+0.2%	0.12	-1
Consumer Staples (XLP)	Fee	20 bps	25 bps	-0.2%	-0.10	-1
Consumer Staples (XLP)	Slippage	0 bps	10 bps	+0.3%	0.23	-1
Consumer Staples (XLP)	Slippage	5 bps	15 bps	+0.2%	0.12	-1
Consumer Staples (XLP)	Slippage	10 bps	20 bps	-0.0%	0.01	-1

SECTOR	STRESS TYPE	SCENARIO	TOTAL COST	HOLDOUT CAGR	HOLDOUT SHARPE	H M
Energy (XLE)	Fee	5 bps	10 bps	+3.3%	0.76	-4
Energy (XLE)	Fee	10 bps	15 bps	+3.1%	0.72	-4
Energy (XLE)	Fee	20 bps	25 bps	+2.7%	0.63	-4
Energy (XLE)	Slippage	0 bps	10 bps	+3.3%	0.76	-4
Energy (XLE)	Slippage	5 bps	15 bps	+3.1%	0.72	-4
Energy (XLE)	Slippage	10 bps	20 bps	+2.9%	0.67	-4
Health Care (XLV)	Fee	5 bps	10 bps	-1.4%	-1.32	-1
Health Care (XLV)	Fee	10 bps	15 bps	-1.5%	-1.32	-1
Health Care (XLV)	Fee	20 bps	25 bps	-1.6%	-1.32	-1
Health Care (XLV)	Slippage	0 bps	10 bps	-1.4%	-1.32	-1
Health Care (XLV)	Slippage	5 bps	15 bps	-1.5%	-1.32	-1
Health Care (XLV)	Slippage	10 bps	20 bps	-1.5%	-1.32	-1
Information Technology (XLK)	Fee	5 bps	10 bps	+6.8%	1.72	+

SECTOR	STRESS TYPE	SCENARIO	TOTAL COST	HOLDOUT CAGR	HOLDOUT SHARPE	H M
Information Technology (XLK)	Fee	10 bps	15 bps	+6.7%	1.71	+
Information Technology (XLK)	Fee	20 bps	25 bps	+6.5%	1.71	+
Information Technology (XLK)	Slippage	0 bps	10 bps	+6.8%	1.72	+
Information Technology (XLK)	Slippage	5 bps	15 bps	+6.7%	1.71	+
Information Technology (XLK)	Slippage	10 bps	20 bps	+6.6%	1.71	+
Manufacturing / Industrials (XLI)	Fee	5 bps	10 bps	+11.6%	2.28	-
Manufacturing / Industrials (XLI)	Fee	10 bps	15 bps	+11.2%	2.24	-
Manufacturing / Industrials (XLI)	Fee	20 bps	25 bps	+10.5%	2.14	-
Manufacturing / Industrials (XLI)	Slippage	0 bps	10 bps	+11.6%	2.28	-

SECTOR	STRESS TYPE	SCENARIO	TOTAL COST	HOLDOUT CAGR	HOLDOUT SHARPE	H M
Manufacturing / Industrials (XLI)	Slippage	5 bps	15 bps	+11.2%	2.24	-1
Manufacturing / Industrials (XLI)	Slippage	10 bps	20 bps	+10.9%	2.19	-1
Retail / Consumer Discretionary (XLY)	Fee	5 bps	10 bps	+5.6%	0.87	-3
Retail / Consumer Discretionary (XLY)	Fee	10 bps	15 bps	+5.0%	0.78	-3
Retail / Consumer Discretionary (XLY)	Fee	20 bps	25 bps	+3.7%	0.61	-3
Retail / Consumer Discretionary (XLY)	Slippage	0 bps	10 bps	+5.6%	0.87	-3
Retail / Consumer Discretionary (XLY)	Slippage	5 bps	15 bps	+5.0%	0.78	-3
Retail / Consumer	Slippage	10 bps	20 bps	+4.3%	0.70	-3

SECTOR	STRESS TYPE	SCENARIO	TOTAL COST	HOLDOUT CAGR	HOLDOUT SHARPE	H M
Discretionary (XLY)						
Utilities (XLU)	Fee	5 bps	10 bps	+ 18.0%	3.04	-3
Utilities (XLU)	Fee	10 bps	15 bps	+ 16.7%	2.87	-3
Utilities (XLU)	Fee	20 bps	25 bps	+ 14.2%	2.51	-3
Utilities (XLU)	Slippage	0 bps	10 bps	+ 18.0%	3.04	-3
Utilities (XLU)	Slippage	5 bps	15 bps	+ 16.7%	2.87	-3
Utilities (XLU)	Slippage	10 bps	20 bps	+ 15.5%	2.69	-3

REGIME AND YEAR

Out-Of-Sample Performance By Regime And Year

These tables use only the averaged ensemble and slice the full out-of-sample stream by macro regime and by calendar year.

SECTOR	REGIME	CAGR	SHARPE	MAX DD	I I
Commodity / Materials (XLB)	Credit Deleveraging	+9.2%	1.47	-16.5%	6

SECTOR	REGIME	CAGR	SHARPE	MAX DD	
Commodity / Materials (XLB)	Disinflationary Growth	+8.6%	0.55	-10.9%	6
Commodity / Materials (XLB)	Fragile Late-Cycle Watch	+6.1%	0.96	-13.6%	6
Commodity / Materials (XLB)	Inflationary Boom	+6.1%	-0.08	-9.2%	5
Commodity / Materials (XLB)	Liquidity Bubble Or Valuation Stretch	+25.3%	0.89	-6.8%	6
Commodity / Materials (XLB)	Panic Or Forced Liquidation	-4.2%	-0.84	-20.0%	5
Commodity / Materials (XLB)	Rate-Shock Regime	+2533.8%	n/a	+0.0%	1
Commodity / Materials (XLB)	Recovery And Reflation	+23.3%	0.86	-4.4%	5
Commodity / Materials (XLB)	Sideways Low- Volatility Regime	+16.1%	1.97	-1.1%	8
Commodity / Materials (XLB)	Unknown	-5.7%	-0.43	-19.3%	4
Consumer Staples (XLP)	Credit Deleveraging	+1.8%	0.88	-25.3%	6
Consumer Staples (XLP)	Disinflationary Growth	+1.1%	-0.14	-22.5%	5
Consumer Staples (XLP)	Fragile Late-Cycle Watch	+1.0%	0.87	-22.1%	6

SECTOR	REGIME	CAGR	SHARPE	MAX DD	
Consumer Staples (XLP)	Inflationary Boom	+1.1%	-0.35	-6.9%	5
Consumer Staples (XLP)	Liquidity Bubble Or Valuation Stretch	+4.1%	0.58	-6.5%	7
Consumer Staples (XLP)	Panic Or Forced Liquidation	-55.3%	-3.03	-16.7%	2
Consumer Staples (XLP)	Rate-Shock Regime	+104.3%	0.12	-4.5%	6
Consumer Staples (XLP)	Recovery And Reflation	-1.2%	1.64	-2.4%	6
Consumer Staples (XLP)	Sideways Low-Volatility Regime	+2.7%	0.65	-5.5%	6
Consumer Staples (XLP)	Unknown	-5.2%	-0.78	-11.2%	4
Energy (XLE)	Credit Deleveraging	+5.1%	0.60	-11.0%	5
Energy (XLE)	Disinflationary Growth	+5.1%	-0.20	-22.5%	5
Energy (XLE)	Fragile Late-Cycle Watch	+4.7%	0.77	-24.6%	6
Energy (XLE)	Inflationary Boom	+4.5%	0.61	-0.4%	6
Energy (XLE)	Liquidity Bubble Or Valuation Stretch	+19.2%	0.43	-4.5%	6

SECTOR	REGIME	CAGR	SHARPE	MAX DD	
Energy (XLE)	Panic Or Forced Liquidation	+173.7%	-0.17	-3.5%	3
Energy (XLE)	Rate-Shock Regime	+1002.2%	1.77	+0.0%	1
Energy (XLE)	Recovery And Reflation	+36.9%	1.44	+0.0%	1
Energy (XLE)	Sideways Low-Volatility Regime	+8.9%	-0.48	-21.4%	5
Energy (XLE)	Unknown	+18.7%	1.21	-9.4%	6
Health Care (XLV)	Credit Deleveraging	+3.9%	0.05	-8.2%	4
Health Care (XLV)	Disinflationary Growth	+3.9%	0.17	-10.9%	6
Health Care (XLV)	Fragile Late-Cycle Watch	+3.5%	0.91	-17.1%	6
Health Care (XLV)	Inflationary Boom	+3.9%	-0.41	-9.7%	4
Health Care (XLV)	Liquidity Bubble Or Valuation Stretch	+14.5%	0.96	-2.6%	1
Health Care (XLV)	Panic Or Forced Liquidation	-34.8%	-0.28	-7.9%	4
Health Care (XLV)	Rate-Shock Regime	+612.8%	0.99	-6.6%	5
Health Care (XLV)	Recovery And Reflation	+1.5%	1.01	-2.5%	5

SECTOR	REGIME	CAGR	SHARPE	MAX DD	
Health Care (XLV)	Sideways Low-Volatility Regime	+6.5%	0.46	-7.3%	5
Health Care (XLV)	Unknown	-3.2%	-0.41	-6.3%	4
Information Technology (XLK)	Credit Deleveraging	+6.9%	1.53	-9.5%	8
Information Technology (XLK)	Disinflationary Growth	+6.4%	1.65	-7.9%	7
Information Technology (XLK)	Fragile Late-Cycle Watch	+5.3%	0.73	-11.8%	6
Information Technology (XLK)	Inflationary Boom	+5.3%	-0.66	-14.1%	4
Information Technology (XLK)	Liquidity Bubble Or Valuation Stretch	+22.6%	1.48	-2.6%	3
Information Technology (XLK)	Panic Or Forced Liquidation	-5.2%	-0.71	-10.8%	3
Information Technology (XLK)	Rate-Shock Regime	+1492.2%	2.16	-1.5%	6
Information Technology (XLK)	Recovery And Reflation	+9.1%	2.04	+0.0%	3
Information Technology (XLK)	Sideways Low-Volatility Regime	+11.7%	1.29	-4.9%	5
Information Technology (XLK)	Unknown	-9.5%	-1.08	-13.1%	4

SECTOR	REGIME	CAGR	SHARPE	MAX DD	
Manufacturing / Industrials (XLI)	Credit Deleveraging	+4.4%	1.43	-33.7%	6
Manufacturing / Industrials (XLI)	Disinflationary Growth	+4.4%	0.33	-20.6%	5
Manufacturing / Industrials (XLI)	Fragile Late-Cycle Watch	+5.1%	1.23	-21.4%	6
Manufacturing / Industrials (XLI)	Inflationary Boom	+4.1%	-0.39	-4.9%	2
Manufacturing / Industrials (XLI)	Liquidity Bubble Or Valuation Stretch	+21.9%	2.25	-3.9%	9
Manufacturing / Industrials (XLI)	Panic Or Forced Liquidation	-69.2%	-3.49	-33.7%	1
Manufacturing / Industrials (XLI)	Rate-Shock Regime	+841.2%	1.77	+0.0%	1
Manufacturing / Industrials (XLI)	Recovery And Reflation	+1.4%	1.77	-5.8%	6
Manufacturing / Industrials (XLI)	Sideways Low-Volatility Regime	+7.3%	0.65	-4.0%	5
Manufacturing / Industrials (XLI)	Unknown	-5.1%	-0.79	-11.2%	4
Retail / Consumer Discretionary (XLY)	Credit Deleveraging	+6.7%	1.88	-20.0%	6
Retail / Consumer Discretionary (XLY)	Disinflationary Growth	+6.6%	0.33	-10.3%	6

SECTOR	REGIME	CAGR	SHARPE	MAX DD	
Retail / Consumer Discretionary (XLY)	Fragile Late-Cycle Watch	+4.4%	1.07	-16.3%	6
Retail / Consumer Discretionary (XLY)	Inflationary Boom	+4.1%	-0.89	-19.0%	3
Retail / Consumer Discretionary (XLY)	Liquidity Bubble Or Valuation Stretch	+18.6%	0.93	-17.7%	5
Retail / Consumer Discretionary (XLY)	Panic Or Forced Liquidation	-71.1%	-1.77	-25.1%	2
Retail / Consumer Discretionary (XLY)	Rate-Shock Regime	+863.3%	0.30	-11.2%	4
Retail / Consumer Discretionary (XLY)	Recovery And Reflation	+8.9%	2.02	-3.9%	6
Retail / Consumer Discretionary (XLY)	Sideways Low-Volatility Regime	+12.0%	1.62	-2.6%	6
Retail / Consumer Discretionary (XLY)	Unknown	-12.1%	-1.22	-18.9%	4
Utilities (XLU)	Credit Deleveraging	-0.7%	n/a	-13.7%	1
Utilities (XLU)	Disinflationary Growth	-0.7%	-0.59	-18.9%	4
Utilities (XLU)	Fragile Late-Cycle Watch	+0.1%	0.50	-15.9%	5
Utilities (XLU)	Inflationary Boom	-0.4%	-0.64	-7.9%	4

SECTOR	REGIME	CAGR	SHARPE	MAX DD	
Utilities (XLU)	Liquidity Bubble Or Valuation Stretch	+2.5%	2.32	-5.7%	7
Utilities (XLU)	Panic Or Forced Liquidation	+15.0%	n/a	+0.0%	1
Utilities (XLU)	Rate-Shock Regime	-6.6%	1.53	-4.7%	6
Utilities (XLU)	Recovery And Reflation	+2.7%	1.44	+0.0%	7
Utilities (XLU)	Sideways Low-Volatility Regime	-1.2%	-0.00	-14.4%	3
Utilities (XLU)	Unknown	-1.1%	-0.08	-9.5%	5

SECTOR	YEAR	TOTAL RETURN	CAGR	SHARPE	MAX DD
Commodity / Materials (XLB)	2005	-9.8%	-9.9%	-0.74	-1
Consumer Staples (XLP)	2005	-6.8%	-6.9%	-1.13	-1
Energy (XLE)	2005	+19.5%	+19.8%	1.16	-9
Health Care (XLV)	2005	+0.9%	+0.9%	0.16	-6
Information Technology (XLK)	2005	-12.3%	-12.5%	-1.25	-1
Manufacturing / Industrials (XLI)	2005	-9.3%	-9.4%	-1.38	-1

SECTOR	YEAR	TOTAL RETURN	CAGR	SHARPE	M D
Retail / Consumer Discretionary (XLY)	2005	-15.6%	-15.8%	-1.42	-1
Utilities (XLU)	2005	-1.5%	-1.5%	-0.09	-9
Commodity / Materials (XLB)	2006	-3.1%	-3.2%	0.95	-4
Consumer Staples (XLP)	2006	-2.3%	-2.3%	0.94	-3
Energy (XLE)	2006	+35.1%	+35.7%	1.84	+
Health Care (XLV)	2006	-3.4%	-3.5%	-0.74	-7
Information Technology (XLK)	2006	-12.3%	-12.5%	n/a	+
Manufacturing / Industrials (XLI)	2006	-7.1%	-7.2%	0.37	-5
Retail / Consumer Discretionary (XLY)	2006	-15.6%	-15.8%	n/a	+
Utilities (XLU)	2006	-2.2%	-2.2%	-1.01	-0
Commodity / Materials (XLB)	2007	+4.2%	+4.3%	0.61	-1
Consumer Staples (XLP)	2007	-6.7%	-6.8%	-0.75	-5
Energy (XLE)	2007	+44.9%	+45.8%	0.76	-8
Health Care (XLV)	2007	-9.1%	-9.2%	-0.60	-8

SECTOR	YEAR	TOTAL RETURN	CAGR	SHARPE	M D
Information Technology (XLK)	2007	-5.3%	-5.3%	0.73	-1
Manufacturing / Industrials (XLI)	2007	-8.2%	-8.3%	-0.06	-1
Retail / Consumer Discretionary (XLY)	2007	-27.6%	-28.0%	-1.33	-1
Utilities (XLU)	2007	+6.1%	+6.2%	0.83	-8
Commodity / Materials (XLB)	2008	+9.6%	+10.6%	0.43	-1
Consumer Staples (XLP)	2008	-20.0%	-21.6%	-1.75	-1
Energy (XLE)	2008	+54.2%	+60.4%	0.56	-1
Health Care (XLV)	2008	-16.8%	-18.1%	-1.54	-8
Information Technology (XLK)	2008	+6.6%	+7.2%	2.28	-0
Manufacturing / Industrials (XLI)	2008	-8.2%	-8.9%	n/a	+
Retail / Consumer Discretionary (XLY)	2008	-20.9%	-22.6%	0.85	-5
Utilities (XLU)	2008	+6.1%	+6.7%	n/a	+
Commodity / Materials (XLB)	2009	+27.9%	+28.1%	0.72	-2
Consumer Staples (XLP)	2009	-17.9%	-18.0%	0.30	-1

SECTOR	YEAR	TOTAL RETURN	CAGR	SHARPE	M D
Energy (XLE)	2009	+65.9%	+66.4%	0.78	-3
Health Care (XLV)	2009	-6.8%	-6.8%	1.13	-6
Information Technology (XLK)	2009	+5.2%	+5.2%	0.02	-1
Manufacturing / Industrials (XLI)	2009	-15.7%	-15.7%	-0.18	-3
Retail / Consumer Discretionary (XLY)	2009	-16.5%	-16.6%	0.37	-2
Utilities (XLU)	2009	+6.1%	+6.1%	n/a	+
Commodity / Materials (XLB)	2010	+49.5%	+50.2%	0.85	-1
Consumer Staples (XLP)	2010	-13.1%	-13.2%	0.71	-7
Energy (XLE)	2010	+75.8%	+77.0%	0.40	-1
Health Care (XLV)	2010	-3.3%	-3.4%	0.37	-1
Information Technology (XLK)	2010	+12.0%	+12.1%	0.48	-9
Manufacturing / Industrials (XLI)	2010	+3.8%	+3.8%	1.21	-1
Retail / Consumer Discretionary (XLY)	2010	+4.6%	+4.7%	1.55	-1
Utilities (XLU)	2010	+2.3%	+2.3%	-0.33	-8

SECTOR	YEAR	TOTAL RETURN	CAGR	SHARPE	M D
Commodity / Materials (XLB)	2011	+71.6%	+72.7%	1.36	-4
Consumer Staples (XLP)	2011	-3.2%	-3.2%	1.59	-3
Energy (XLE)	2011	+118.0%	+120.0%	1.70	-5
Health Care (XLV)	2011	+3.4%	+3.5%	0.58	-1
Information Technology (XLK)	2011	+20.8%	+21.1%	1.78	+
Manufacturing / Industrials (XLI)	2011	+2.5%	+2.5%	-0.12	-8
Retail / Consumer Discretionary (XLY)	2011	+19.3%	+19.6%	1.70	-4
Utilities (XLU)	2011	+7.8%	+7.9%	1.53	+
Commodity / Materials (XLB)	2012	+95.2%	+96.0%	1.66	-3
Consumer Staples (XLP)	2012	+3.1%	+3.1%	2.10	-1
Energy (XLE)	2012	+115.5%	+116.5%	-0.11	-6
Health Care (XLV)	2012	+12.9%	+12.9%	1.77	-2
Information Technology (XLK)	2012	+29.1%	+29.3%	1.38	-2
Manufacturing / Industrials (XLI)	2012	+9.3%	+9.4%	1.06	-4

SECTOR	YEAR	TOTAL RETURN	CAGR	SHARPE	M D
Retail / Consumer Discretionary (XLY)	2012	+38.2%	+38.5%	2.72	-1
Utilities (XLU)	2012	+5.7%	+5.7%	-1.30	-1
Commodity / Materials (XLB)	2013	+110.6%	+111.5%	0.75	-5
Consumer Staples (XLP)	2013	+13.1%	+13.1%	1.92	-1
Energy (XLE)	2013	+129.9%	+131.1%	0.93	-4
Health Care (XLV)	2013	+37.9%	+38.2%	2.57	-5
Information Technology (XLK)	2013	+40.8%	+41.1%	1.26	-2
Manufacturing / Industrials (XLI)	2013	+25.6%	+25.8%	1.47	-4
Retail / Consumer Discretionary (XLY)	2013	+67.8%	+68.3%	2.55	-4
Utilities (XLU)	2013	+4.2%	+4.2%	-0.27	-5
Commodity / Materials (XLB)	2014	+117.3%	+118.3%	0.85	-3
Consumer Staples (XLP)	2014	+12.4%	+12.5%	-0.11	-3
Energy (XLE)	2014	+86.5%	+87.3%	-1.68	-2
Health Care (XLV)	2014	+57.5%	+58.0%	1.96	-4

SECTOR	YEAR	TOTAL RETURN	CAGR	SHARPE	M D
Information Technology (XLK)	2014	+50.6%	+51.0%	1.19	-3
Manufacturing / Industrials (XLI)	2014	+30.1%	+30.3%	0.48	-9
Retail / Consumer Discretionary (XLY)	2014	+71.5%	+72.1%	0.39	-5
Utilities (XLU)	2014	+5.3%	+5.3%	0.23	-6
Commodity / Materials (XLB)	2015	+136.2%	+157.5%	1.57	-2
Consumer Staples (XLP)	2015	+11.8%	+13.1%	-0.29	-1
Energy (XLE)	2015	+76.5%	+86.8%	-1.09	-7
Health Care (XLV)	2015	+60.4%	+68.2%	0.29	-5
Information Technology (XLK)	2015	+51.8%	+58.3%	0.20	-4
Manufacturing / Industrials (XLI)	2015	+32.2%	+36.0%	0.48	-3
Retail / Consumer Discretionary (XLY)	2015	+76.0%	+86.3%	1.34	-0
Utilities (XLU)	2015	-1.2%	-1.3%	-1.64	-6
Commodity / Materials (XLB)	2016	+146.7%	+166.6%	1.62	+
Consumer Staples (XLP)	2016	+13.2%	+14.4%	0.24	-5

SECTOR	YEAR	TOTAL RETURN	CAGR	SHARPE	M D
Energy (XLE)	2016	+82.9%	+92.7%	1.04	+
Health Care (XLV)	2016	+53.2%	+59.0%	-0.40	-8
Information Technology (XLK)	2016	+71.3%	+79.5%	1.83	-2
Manufacturing / Industrials (XLI)	2016	+40.7%	+45.0%	1.18	-0
Retail / Consumer Discretionary (XLY)	2016	+94.1%	+105.5%	1.18	-6
Utilities (XLU)	2016	-1.2%	-1.3%	-1.04	-0
Commodity / Materials (XLB)	2017	+158.7%	+161.7%	0.84	-3
Consumer Staples (XLP)	2017	+12.6%	+12.7%	-0.07	-5
Energy (XLE)	2017	+87.2%	+88.6%	0.30	-5
Health Care (XLV)	2017	+60.7%	+61.6%	1.75	-0
Information Technology (XLK)	2017	+102.4%	+104.1%	2.08	-4
Manufacturing / Industrials (XLI)	2017	+60.4%	+61.3%	2.28	-2
Retail / Consumer Discretionary (XLY)	2017	+103.8%	+105.6%	1.40	-2
Utilities (XLU)	2017	-1.2%	-1.2%	0.03	-7

SECTOR	YEAR	TOTAL RETURN	CAGR	SHARPE	M D
Commodity / Materials (XLB)	2018	+181.6%	+248.4%	2.05	+
Consumer Staples (XLP)	2018	+14.2%	+17.4%	0.36	-5
Energy (XLE)	2018	+85.4%	+110.5%	-0.22	-4
Health Care (XLV)	2018	+60.9%	+77.4%	0.07	-6
Information Technology (XLK)	2018	+117.5%	+155.2%	1.82	+
Manufacturing / Industrials (XLI)	2018	+71.1%	+91.1%	1.79	+
Retail / Consumer Discretionary (XLY)	2018	+121.9%	+161.4%	2.62	+
Utilities (XLU)	2018	-8.4%	-10.0%	-1.32	-9
Commodity / Materials (XLB)	2019	+197.7%	+548.9%	1.50	-0
Consumer Staples (XLP)	2019	+24.0%	+44.6%	2.30	-1
Energy (XLE)	2019	+89.7%	+199.7%	1.42	-0
Health Care (XLV)	2019	+58.4%	+120.1%	-0.24	-6
Information Technology (XLK)	2019	+128.5%	+312.3%	1.01	-4
Manufacturing / Industrials (XLI)	2019	+70.8%	+150.3%	-0.03	-4

SECTOR	YEAR	TOTAL RETURN	CAGR	SHARPE	M D
Retail / Consumer Discretionary (XLY)	2019	+122.7%	+294.6%	0.12	-3
Utilities (XLU)	2019	-8.4%	-14.0%	-0.00	-2
Commodity / Materials (XLB)	2020	+201.2%	+331.7%	1.15	+
Consumer Staples (XLP)	2020	+24.0%	+33.0%	n/a	+
Energy (XLE)	2020	+93.6%	+140.2%	0.35	-5
Health Care (XLV)	2020	+96.4%	+144.7%	1.85	-5
Information Technology (XLK)	2020	+128.5%	+199.2%	n/a	+
Manufacturing / Industrials (XLI)	2020	+90.7%	+135.5%	2.11	+
Retail / Consumer Discretionary (XLY)	2020	+120.7%	+185.8%	-0.58	-1
Utilities (XLU)	2020	-8.0%	-10.5%	1.15	+
Commodity / Materials (XLB)	2021	+209.6%	+212.5%	1.41	+
Consumer Staples (XLP)	2021	+29.5%	+29.7%	0.99	-3
Energy (XLE)	2021	+107.1%	+108.3%	1.01	+
Health Care (XLV)	2021	+99.0%	+100.1%	0.45	-2

SECTOR	YEAR	TOTAL RETURN	CAGR	SHARPE	M D
Information Technology (XLK)	2021	+159.8%	+161.7%	1.77	-2
Manufacturing / Industrials (XLI)	2021	+112.9%	+114.2%	2.06	-0
Retail / Consumer Discretionary (XLY)	2021	+148.5%	+150.3%	0.92	-8
Utilities (XLU)	2021	-1.5%	-1.5%	1.61	+
Commodity / Materials (XLB)	2022	+192.9%	+196.6%	-1.00	-5
Consumer Staples (XLP)	2022	+25.3%	+25.7%	-0.42	-5
Energy (XLE)	2022	+120.0%	+122.1%	1.01	+
Health Care (XLV)	2022	+102.1%	+103.8%	0.20	-8
Information Technology (XLK)	2022	+150.4%	+153.1%	-0.26	-9
Manufacturing / Industrials (XLI)	2022	+108.9%	+110.7%	-0.36	-4
Retail / Consumer Discretionary (XLY)	2022	+110.0%	+111.9%	-0.91	-1
Utilities (XLU)	2022	-3.7%	-3.7%	-0.18	-1
Commodity / Materials (XLB)	2023	+189.0%	+193.5%	-0.27	-4
Consumer Staples (XLP)	2023	+21.6%	+21.9%	-0.74	-4

SECTOR	YEAR	TOTAL RETURN	CAGR	SHARPE	M D
Energy (XLE)	2023	+121.2%	+123.8%	1.06	+
Health Care (XLV)	2023	+95.5%	+97.4%	-1.26	-3
Information Technology (XLK)	2023	+146.6%	+149.9%	-1.06	-1
Manufacturing / Industrials (XLI)	2023	+104.6%	+106.7%	-1.06	-2
Retail / Consumer Discretionary (XLY)	2023	+101.3%	+103.4%	-1.06	-4
Utilities (XLU)	2023	-5.0%	-5.1%	-0.19	-5
Commodity / Materials (XLB)	2024	+195.6%	+196.7%	1.06	-0
Consumer Staples (XLP)	2024	+21.6%	+21.6%	n/a	+
Energy (XLE)	2024	+128.6%	+129.3%	1.21	-0
Health Care (XLV)	2024	+97.9%	+98.3%	1.10	+
Information Technology (XLK)	2024	+153.1%	+153.9%	1.10	+
Manufacturing / Industrials (XLI)	2024	+133.7%	+134.4%	2.05	+
Retail / Consumer Discretionary (XLY)	2024	+118.4%	+119.0%	1.81	+
Utilities (XLU)	2024	-5.5%	-5.5%	-0.18	-2

SECTOR	YEAR	TOTAL RETURN	CAGR	SHARPE	M D
Commodity / Materials (XLB)	2025	+204.1%	+206.2%	0.87	-1
Consumer Staples (XLP)	2025	+19.9%	+20.0%	-1.18	-1
Energy (XLE)	2025	+136.9%	+138.2%	0.84	-4
Health Care (XLV)	2025	+94.5%	+95.3%	-1.56	-1
Information Technology (XLK)	2025	+172.9%	+174.6%	2.03	+
Manufacturing / Industrials (XLI)	2025	+168.0%	+169.6%	2.98	+
Retail / Consumer Discretionary (XLY)	2025	+138.3%	+139.6%	1.41	-3
Utilities (XLU)	2025	+4.8%	+4.8%	2.46	-3
Commodity / Materials (XLB)	2026	+204.1%	+87011.7%	n/a	+
Consumer Staples (XLP)	2026	+21.8%	+231.9%	2.51	+
Energy (XLE)	2026	+136.9%	+18967.7%	n/a	+
Health Care (XLV)	2026	+94.5%	+5630.9%	n/a	+
Information Technology (XLK)	2026	+172.9%	+44971.8%	n/a	+
Manufacturing / Industrials (XLI)	2026	+164.6%	+37259.8%	-2.09	-1

SECTOR	YEAR	TOTAL RETURN	CAGR	SHARPE	M D
Retail / Consumer Discretionary (XLY)	2026	+131.1%	+16295.2%	-3.63	-3
Utilities (XLU)	2026	+13.2%	+112.2%	3.77	-0